A Consistent Independence Test via Projected Mutual Information

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Abstract

In this paper, we propose a nonparametric independence test based on mutual information. Distinguished from the previous work, we estimate the mutual information in a conditional density form, whose dimension could be reduced to 1 with new projection methods.The optimal projection direction is estimated by maximizing a penalized mutual information. Based on the optimal projection, we construct an independence test via the projected mutual information, which is insensitive to the dimensions of random vectors. The test is consistent against fixed alternatives, and can detect local alternatives at a fast rate as if the variables were univariate. Numerical results indicate that the test is more powerful compared with other existing independence tests, especially when the sample size is small or the dimension is large.

KEY WORDS: Independence test, Mutual information, Projection Pursuit, Local alternative.

1. INTRODUCTION

Independence is one of the fundamental concepts in data analysis and statistical inference. Suppose $\mathbf{x} = (X_1, \ldots, X_p)^{\mathrm{T}} \in \mathbb{R}^p$ and $\mathbf{y} = (Y_1, \ldots, Y_q)^{\mathrm{T}} \in \mathbb{R}^q$ are two random vectors. The null hypothesis states that the two random vectors are independent, namely, $H_0 : \mathbf{x} \perp\!\!\!\perp \mathbf{y}$. Numerous efforts have been made to measure complex dependence between random vectors where the resultant tests are consistent against any fixed alternatives. Gretton et al. (2005) introduced a Hilbert-Schmidt independence criterion using an empirical estimate of the Hilbert-Schmidt norm of the cross-covariance operator. Székely et al. (2007) proposed the distance correlation to characterize dependence between two random vectors of arbitrary dimension without distribution assumptions, provided the existence of their moments. Extension of this concept has been discussed among high dimensional covariates (Székely and Rizzo, 2013; Yao et al., 2018), fast computations (Huo and Székely, 2016) and so on. Horváth and Rice (2015) proposed a test based on norms of empirical cross covariance operators to test independence between functional time series. Zhu et al. (2017) extended the Hoeffding coefficient (Hoeffding, 1948) via projection pursuit and proposed the projection correlation, which removed the moment constraints of distance correlation. Testing independence is also closely related to testing mutual independence and conditional independence, see, for example, Matilla-García and Marín (2008); Su and White (2014).

Mutual Information (Shannon, 1948), which is defined as

$$I(\mathbf{x}, \mathbf{y}) = E\left\{\log \frac{f_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})}{f_{\mathbf{x}}(\mathbf{x}) f_{\mathbf{y}}(\mathbf{y})}\right\}, \tag{1.1}$$

where $f_{\mathbf{x}}(\mathbf{x})$, $f_{\mathbf{y}}(\mathbf{y})$, $f_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})$ are the density functions of $\mathbf{x}$, $\mathbf{y}$ and $(\mathbf{x}, \mathbf{y})$, respectively, is one of the most popular measures of dependence between two random vectors. Unlike

Pearson correlation which only captures the linear dependence, mutual information is also sensitive to nonlinear associations. It is symmetric, nonnegative and equals zero if and only if $\mathbf{x}$ and $\mathbf{y}$ are statistically independent. Due to its information theory background, mutual information enjoys several appealing properties, such as invariance with respect to reparametrizations. Specifically, let $\mathbf{x}' = \mathbf{T}_1(\mathbf{x})$ and $\mathbf{y}' = \mathbf{T}_2(\mathbf{y})$ be smooth and uniquely invertible transformations of $\mathbf{x}$ and $\mathbf{y}$, then $I(\mathbf{x}', \mathbf{y}') = I(\mathbf{x}, \mathbf{y})$ (Berrett and Samworth, 2019). Moreover, the mutual information provides an upper bound on how well any test of dependence can perform since it is defined as the log-likelihood ratio (Kinney and Atwal, 2014). With these attractive properties, mutual information is always regarded as a powerful measure of nonlinear dependence, and has received considerable attention in the literature. In terms of estimation, the k-nearest neighbor estimator was proposed by Singh et al. (2003), and the density functions with nonparametric models was proposed by Peter and Rangarajan (2008). Recently, Zeng et al. (2018) considered a jackknife version of the kernel estimate with equalized bandwidth in estimating mutual information. Moreover, Berrett and Samworth (2019) proposed the MINT, i.e., mutual information nonparametric test, where the mutual information is obtained by efficient entropy estimators.

We next briefly summarize the motivations and contributions of the current paper. Essentially, the mutual information has to be estimated nonparametrically when the distributions of $\mathbf{x}$ and $\mathbf{y}$ are unknown. However, due to the well known curse of dimensionality, the performance of nonparametric estimators deteriorate quickly as the dimension increases. Directly implementing nonparametric estimates in independence tests with high dimensional random vectors may have little power (Ramdas et al., 2015; Guo and Zhu, 2017). This motivates us to reduce the dimension through a projection pursuit approach. Consider the scenario where $\mathbf{x}$ is of large dimension. Instead

of directly estimating $I(\mathbf{x}, \mathbf{y})$, we choose to estimate $\sup_{\|\boldsymbol{\beta}\|=1} I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y})$, which is the supremum mutual information between $\mathbf{y}$ and the projection of $\mathbf{x}$ onto a unit vector. Define $\boldsymbol{\beta}_0$ as the unit vector that maximize $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y})$. Then one may use $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0, \mathbf{y})$ to quantify the dependence between $\mathbf{x}$ and $\mathbf{y}$, in view of the fact that $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0, \mathbf{y}) = 0$ under the null and $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0, \mathbf{y}) > 0$ under the alternative hypothesis. However, when $\mathbf{x}$ and $\mathbf{y}$ are independent, $\boldsymbol{\beta}_0$ is unidentifiable simply because $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y}) = 0$ holds for any $\boldsymbol{\beta}$. Consequently, the asymptotic distribution of the projected mutual information becomes intractable. We solve this challenge by estimating $\boldsymbol{\beta}_0$ through the penalized mutual information (2.1). We carefully study the asymptotic distribution of $\widehat{\boldsymbol{\beta}}_0$, and construct the independence test based on the projected mutual information, whose distribution converges to a normal distribution at rate $(nh_x^{1/2}h_y^{1/2})$ under the null hypothesis, where h_x and h_y are the kernel bandwidths defined in (2.2). Under the fixed alternative hypothesis, the test statistic converges to normal at rate $n^{1/2}$. The test has asymptotic power one for local alternatives as long as its distance from null is larger than $(n^{-1/2}h_x^{-1/4}h_y^{-1/4}\log n)$, which is discussed in detail in Section 2.3. In summary, we make the following contributions in this paper:

1. We propose a penalized mutual information to estimate $\boldsymbol{\beta}_0$. Under mild conditions, the estimator converges to a stationary vector under the null hypothesis, and is root-n consistent under the alternative hypothesis.

2. The proposed test statistic behaves almost as if the model was univariate. The convergence rate of the statistic does not depend on the dimension of the $\mathbf{x}$. Asymptotically, the curse of dimensionality is weakened.

3. The proposed independence test is consistent against all fixed dependent alternatives. It is also consistent against local alternatives converging to the null at rate arbitrarily slower than $n^{1/2}h_x^{1/4}h_y^{1/4}/\log n$.

This paper is organized as follows. We propose the penalized mutual information in Section 2.1, and implement a fast computation algorithm to estimate $\widehat{\boldsymbol{\beta}}$ in Section 2.2. We study the asymptotic properties of the proposed projected mutual information test in detail in Section 2.3. In Section 3, we extend the test to the case where both $\mathbf{x}$ and $\mathbf{y}$ have relatively large dimensions. Extensive numerical analysis that illustrate the usefulness of the proposed method are carried out in Section 4. Some concluding remarks are given in Section 5. All technical proofs are relegated to the Appendix.

2. INDEPENDENCE TEST WITH PENALIZED MUTUAL INFORMATION

2.1. Penalized Mutual Information

In this section, we discuss how to estimate the maximum unit direction by implementing the penalized mutual information. Assume that the dimension of Y is one. We first give a formal definition for the maximum unit direction $\boldsymbol{\beta}_0$ when $\mathbf{x}$ and Y are dependent. Specifically, $\boldsymbol{\beta}_0$ is defined as

$$\boldsymbol{\beta}_0 \stackrel{\text{def}}{=} \arg\max_{\|\boldsymbol{\beta}\|=1} \left\{ I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y) \right\}.$$

We further assume that the first element of $\boldsymbol{\beta}_0$ is positive. Because mutual information characterizes the dependence between $\mathbf{x}$ and Y, if we estimate $\boldsymbol{\beta}_0$ accurately, a test based on $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0, Y)$ should be consistent. We illustrate this idea formally in Lemma 1, which indicates that an independence test based on the maximal projected mutual information should be consistent.

Lemma 1. $\mathbf{x}$ *and* Y *are independent if and only if*

$$\sup_{\|\boldsymbol{\beta}\|=1} I(\mathbf{x}^{T}\boldsymbol{\beta}, Y) = 0.$$

The proof Lemma 1 is given in Appendix A1. Based on this lemma, one may directly maximize $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y)$, obtain estimator $\check{\boldsymbol{\beta}}$, then develop an independence test based on $I(\mathbf{x}^{\mathrm{T}}\check{\boldsymbol{\beta}}, Y)$. However, under the null hypothesis, $\check{\boldsymbol{\beta}}$ is not identifiable and its asymptotic null distributions becomes intractable. To achieve identification, we add a penalty term to the mutual information to guarantee that the estimator is obtained at a pre-specified value $\boldsymbol{\beta}^*$ under the null hypothesis. Specifically, define

$$\widehat{\boldsymbol{\beta}} = \arg\max_{\|\boldsymbol{\beta}\|=1} \left\{ U_n(\boldsymbol{\beta}) - c_n \mathbb{I}(\boldsymbol{\beta} \neq \boldsymbol{\beta}^*) \right\}, \tag{2.1}$$

where $\boldsymbol{\beta}^*$ is an arbitrary p dimensional vector such that $\|\boldsymbol{\beta}^*\| = 1$ and $\mathbb{I}(\cdot)$ is an indicator function. c_n is a sequence of positive real numbers decreasing to zero at an appropriate rate. One may simply let $U_n(\boldsymbol{\beta})$ be $\widehat{I}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y)$, the nonparametric kernel estimator of $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y)$. However, the bias caused by the nonparametric estimate would reduce the test's power under local alternatives. This motivates us to consider an unbiased estimator of $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y)$. Formally, we let

$$U_n(\boldsymbol{\beta}) \stackrel{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n} \log\left\{ \frac{\widehat{f}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta},Y}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)}{\widehat{f}_Y(Y_i)\widehat{f}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})} \right\} + \frac{\nu_2^2}{2nh_xh_y}\widehat{D}_1(\boldsymbol{\beta}) - \frac{\nu_2}{2nh_x}\widehat{D}_2(\boldsymbol{\beta}) - \frac{\nu_2}{2nh_y}\widehat{D}_3,$$

where $\widehat{D}_1(\boldsymbol{\beta}) = n^{-1}\sum_{i=1}^{n} \widehat{f}^{-1}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta},Y}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)$, $\widehat{D}_2(\boldsymbol{\beta}) = n^{-1}\sum_{i=1}^{n} \widehat{f}^{-1}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})$, and $\widehat{D}_3 = n^{-1}\sum_{i=1}^{n} \widehat{f}_Y^{-1}(Y_i)$, $\nu_2 \stackrel{\text{def}}{=} \int K^2(u)du$, $\widehat{f}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta},Y}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)$, and $\widehat{f}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})$ and $\widehat{f}_Y(Y_i)$ are the leave-one-out kernel density estimators. Specifically,

$$\widehat{f}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta},Y}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i) = (n-1)^{-1}\sum_{j\neq i}^{n} K_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}) K_{h_y}(Y_i - Y_j),$$

$$\widehat{f}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}) = (n-1)^{-1}\sum_{j\neq i}^{n} K_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}), \quad \widehat{f}_Y(Y_i) = \frac{1}{n-1}\sum_{j\neq i}^{n} K_{h_y}(Y_i - Y_j). \tag{2.2}$$

$K_{h_x}(\cdot) = K(\cdot/h_x)/h_x, K_{h_y}(\cdot) = K(\cdot/h_y)/h_y$, where $K(\cdot)$ is a kernel function, h_x and h_y

are bandwidths.

We briefly discuss some intuitions behind the objective function (2.1). Under the null hypothesis, when c_n is large enough compared with $\sup_{\|\boldsymbol{\beta}\|=1} U_n(\boldsymbol{\beta})$, the maximum of (2.1) will be attained at $\boldsymbol{\beta}^*$. Under the alternative, $\widehat{\boldsymbol{\beta}}$ will converge in probability to $\boldsymbol{\beta}_0$ when c_n is negligible. The details on how to choose c_n and $\boldsymbol{\beta}^*$ will be given in Section 4. Next, we carefully study the asymptotic behaviors of $\widehat{\boldsymbol{\beta}}$ under both the null and the alternative hypothesis. The following technical conditions are imposed.

(C1) *(Kernel Function)* The univariate kernel function $K(\cdot)$ is bounded, symmetric about zero and Lipschitz continuous. In addition, it satisfies

$$\int K(\upsilon)d\upsilon = 1, \quad \int \upsilon^i K(\upsilon)d\upsilon = 0, 1 \leq i \leq m-1, \quad 0 \neq \int \upsilon^m K(\upsilon)d\upsilon < \infty.$$

Here m is the order of the kernel function.

(C2) *(Bandwidth)* The bandwidths h_x and h_y satisfy $h_x^{m-1/2}h_y^{-1/2} \to 0$, $h_x^{-1/2}h_y^{m-1/2} \to 0$, $nh_x^{1/2}h_y^{1/2}(h_x^{2m-1/2} + h_y^{2m-1/2})(h_x^{1/2} + h_y^{1/2}) \to 0$, and $nh_x^2h_y^2 \to \infty$.

(C3) *(Density)* The probability density function of $(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta})$, Y and $(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y)$ denoted by $f_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta})$, $f_Y(y)$ and $f_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta},Y}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y)$ are bounded away from 0 to infinity, and have bounded supports.

(C4) *(Derivatives)* The m-th and $(m-1)$-th derivatives of $f_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta})$, $f_Y(Y)$ and $f_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta},Y}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y)$ are locally Lipschitz-continuous with respect to $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}$ and Y.

(C5) *(Conditional Density)* The third derivative of $\log\left\{f_{Y|\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(Y \mid \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta})\right\}$ with respect to $\boldsymbol{\beta}$ around $\boldsymbol{\beta}_0$ can be bounded with a function of $(\mathbf{x}, Y)$ with finite second moment.

These conditions are mild and commonly assumed. Condition (C1) assumes that the order of the kernel function is m. When using the usual kernel functions such as Gaussian and Epanechnikov kernels, $m = 2$. If we let $h_x = h_y = h$, the assumption in (C2) implies $nh^5 \to 0$ and $nh^4 \to \infty$. Condition (C3) assumes that the support is bounded, and is widely used in the literature, for instance, Fan and Jiang (2005), Zhu and Xue (2006), and Su and White (2014). However, the notorious problem of boundary effect may arise when estimating the density functions at boundaries. Numerous methods have been proposed to address this issue, see, for example, Jones (1993) and Hall et al. (2002). One may also refer to Kheireddine (2016) for a nice summary about boundary corrections in kernel density estimation. When the bounded support assumption is not satisfied, by noting that $\mathbf{x}$ and Y are independent if and only if their marginal monotonic transformation counterparts are independent, we may follow Zeng et al. (2018) to consider uniform transformation of $\mathbf{x}$ and Y. Condition (C4) is concerned with the smoothness of the density function. The last condition, (C5), is used to guarantee strong identification of $\boldsymbol{\beta}_0$ when $\boldsymbol{\beta}_0$ is assumed to be unique. This condition is also used in Ma and Zhu (2013).

The following theorem indicates that with an appropriate chosen c_n, $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ holds with probability tending to 1 under the null hypothesis.

Theorem 1. *Assume the regularity conditions (C1)-(C4) hold. Under the null hypothesis,* $\sup_{\|\boldsymbol{\beta}\|=1} U_n(\boldsymbol{\beta}) = O_p(n^{-1}h_x^{-1/2}h_y^{-1/2}\log n)$. *Furthermore, when* $c_n n h_x^{1/2} h_y^{1/2}/\log n \to \infty$, $Pr(\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*) \to 1$.

Under the alternative hypothesis, the penalty term is asymptotically negligible. We prove the asymptotic normality for $\widehat{\boldsymbol{\beta}}$ in Theorem 2.

Theorem 2. *Assume the maximum unit direction* $\boldsymbol{\beta}_0$ *is unique and (C1)-(C5) hold,*

under the alternative hypothesis, when $c_n n^{1/2} \to 0$,

$$n^{1/2}\mathbf{B}_1(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N\left(\mathbf{0}, \mathbf{B}_2\right),$$

where $\mathbf{B}_1$ *and* $\mathbf{B}_2$ *are defines as*

$$\begin{aligned}
\mathbf{B}_1 &= (\mathbf{I}_p - \boldsymbol{\beta}_0\boldsymbol{\beta}_0^T)\, E\left[\frac{\partial^2 \log\left\{f(Y \mid \mathbf{x}^T\boldsymbol{\beta}_0)\right\}}{\partial\boldsymbol{\beta}_0\partial\boldsymbol{\beta}_0^T}\right], \\
\mathbf{B}_2 &= (\mathbf{I}_p - \boldsymbol{\beta}_0\boldsymbol{\beta}_0^T)\, E\Bigg(\left\{\mathbf{x} - E(\mathbf{x} \mid \mathbf{x}^T\boldsymbol{\beta}_0)\right\}\left[\frac{\partial \log\left\{f(Y \mid \mathbf{x}^T\boldsymbol{\beta}_0)\right\}}{\partial\mathbf{x}^T\boldsymbol{\beta}_0}\right]^2 \\
&\qquad \left\{\mathbf{x} - E(\mathbf{x} \mid \mathbf{x}^T\boldsymbol{\beta}_0)\right\}^T\Bigg)(\mathbf{I}_p - \boldsymbol{\beta}_0\boldsymbol{\beta}_0^T).
\end{aligned}$$

These two Theorems show that the estimator $\widehat{\boldsymbol{\beta}}$ is inherently adaptive. Under the null hypothesis, it will converge to the pre-specified direction $\boldsymbol{\beta}^*$. While under the alternative, it will converge to the maximum unit direction $\boldsymbol{\beta}_0$.

To achieve identifiability, we assume that both first elements of $\boldsymbol{\beta}_0$ and $\widehat{\boldsymbol{\beta}}$ are positive. If the first element of $\boldsymbol{\beta}_0$ is zero, we can rotate the covariates $\mathbf{x}$ to ensure that the first element of $\boldsymbol{\beta}_0$ is nonzero. We constrain the parameters on the unit sphere simply because $I(c\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y) = I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y)$ for all $c \neq 0$. The constraint on the first element of the parameter vector is also commonly assumed in the single index model literature (Balabdaoui et al., 2019). If the algorithm outputs a $\widehat{\boldsymbol{\beta}}$ whose first element is negative, we will flip signs of $\widehat{\boldsymbol{\beta}}$. Essentially, when $\boldsymbol{\beta}_0$ is unique, these identifiability constraints on $\widehat{\boldsymbol{\beta}}$ are only used for studying its asymptotic properties under the alternative. We will illustrate in Theorem 4 that the test is still consistent even when the uniqueness assumption is violated. Next, we offer another perspective to understand the uniqueness assumption.

Lemma 2. *(Yin and Cook, 2005) Assume the support of* $\mathbf{x}$ *is a convex set, under the*

general single index regression model $Y = g(\mathbf{x}^T\boldsymbol{\beta}_0, \varepsilon)$*, where the error* ε *is independent of* $\mathbf{x}$*, we have* $I(\mathbf{x}, Y) = I(\mathbf{x}^T\boldsymbol{\beta}_0, Y) \geq I(\mathbf{x}^T\boldsymbol{\beta}, Y)$*, with equality if and only if* $\boldsymbol{\beta}$ *is proportional to* $\boldsymbol{\beta}_0$*.*

Lemma 2 indicates that, when Y is related to $\mathbf{x}$ only through a single index structure, the maximum unit direction $\boldsymbol{\beta}_0$ is proportional to the index direction. Moreover, the direction $\boldsymbol{\beta}_0$ is unique (up to ± 1). However, assuming such single index structure relationship may be too restrictive. Under a more general multi-index structure, i.e., Y depends on $\mathbf{x}$ through more than one linear combinations (at most p), as discussed in Section 3.2 of Yin et al. (2008), the maximum unit direction $\boldsymbol{\beta}_0$ lies in the subspace of the space spanned by all the indexes. More importantly, the direction is unique under mild assumptions (Yin et al., 2008, Proposition 4).

2.2. A Fast Algorithm under Orthogonal Constraint

In this section, we briefly discuss the algorithm for computing $\widehat{\boldsymbol{\beta}}$. We propose to estimate the maximum unit direction by using an advanced Stiefel manifold optimization algorithm proposed in Wen and Yin (2013). The algorithm applies the Cayley transform to preserve the orthogonal constraints and develops a curvilinear search algorithm to obtain a feasible solution. Specifically, let $\mathbf{G} = \partial U_n(\boldsymbol{\beta})/\partial\boldsymbol{\beta}$ be the gradient. Define a skew-symmetric matrix $\mathbf{A}$ as $\mathbf{A} = \mathbf{G}\boldsymbol{\beta}^{\mathrm{T}} - \boldsymbol{\beta}\mathbf{G}^{\mathrm{T}}$ and the next step estimate as $\boldsymbol{\beta}(\tau)$, where τ is the step size. Then $\boldsymbol{\beta}(\tau)$ is given in the closed form

$$\boldsymbol{\beta}(\tau) = \left(\mathbf{I} + \frac{\tau}{2}\mathbf{A}\right)^{-1}\left(\mathbf{I} - \frac{\tau}{2}\mathbf{A}\right)\boldsymbol{\beta}. \tag{2.3}$$

The step size τ can be found by using the nonmonotone line search with the Barzilai-Borwein step size. The update (2.3) preserves the orthogonal constraint elegantly, i.e., $\|\boldsymbol{\beta}(\tau)\| = \|\boldsymbol{\beta}\| = 1$. For more technical details, we refer the readers to Wen and

Yin (2013) and Barzilai and Borwein (1988). Since this is a non-convex problem in nature, the convergence of the algorithm to the global maximum is not guaranteed. Moreover, it is better to choose an initial value that is sufficiently close to $\boldsymbol{\beta}_0$ to obtain a relatively larger test statistic when the true data generation process (dgp) is under the alternative. However, it is not known as a prior that whether the true dgp is under the null or the alternative. Hence the ideal initial value may be impossible to obtain. Yet, our limited numerical experience indicates that the algorithm is not very sensitive to the initial value and performs surprisingly well even with random initializations that are uniformly distributed on the unit sphere. Since the indicator function does not have derivatives, and it equals 0 when $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ while equals 1 otherwise, we first maximize $U_n(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ to obtain $\boldsymbol{\beta}_s$, then compare $U_n(\boldsymbol{\beta}_s) - c_n\mathbb{I}(\boldsymbol{\beta}_s \neq \boldsymbol{\beta}^*)$ and $U_n(\boldsymbol{\beta}^*)$. We set $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}_s$ if $U_n(\boldsymbol{\beta}_s) - c_n\mathbb{I}(\boldsymbol{\beta}_s \neq \boldsymbol{\beta}^*) > U_n(\boldsymbol{\beta}^*)$ and set $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ otherwise.

Algorithm 1 Estimating the maximum unit direction with orthogonal constraint

1. Input error control ϵ, maximum iteration S, and an initial value $\boldsymbol{\beta}_0$.
2. While $1 \leq s \leq S$
   (1) Compute the current gradient matrix $\mathbf{G}$ of $U_n(\boldsymbol{\beta})$ at $\boldsymbol{\beta}_{s-1}$.
   (2) Compute the skew-symmetric matrix $\mathbf{A} = \mathbf{G}\boldsymbol{\beta}_{s-1}^{\mathrm{T}} - \boldsymbol{\beta}_{s-1}\mathbf{G}^{\mathrm{T}}$.
   (3) Compute the Barzilai-Borwein step size τ with nonmonotone line search.
   (4) Set $\boldsymbol{\beta}_s = \boldsymbol{\beta}_{s-1}(\tau)$, where $\boldsymbol{\beta}_{s-1}(\tau)$ is given by (2.3).
   (5) Break if $\|\boldsymbol{\beta}_s - \boldsymbol{\beta}_{s-1}\|_2 \leq \epsilon$.
3. Output $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}_s$ if $U_n(\boldsymbol{\beta}_s) - c_n\mathbb{I}(\boldsymbol{\beta}_s \neq \boldsymbol{\beta}^*) > U_n(\boldsymbol{\beta}^*)$ and $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ otherwise.

2.3. An Independence Test via Projection

After obtaining $\widehat{\boldsymbol{\beta}}$, the test statistic is constructed by the projected mutual information

as follows

$$T_n = U_n(\widehat{\boldsymbol{\beta}}) = \frac{1}{n}\sum_{i=1}^{n} \log\left\{\frac{\widehat{f}_{Y|\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}})}{\widehat{f}_Y(Y_i)}\right\} + \frac{\nu_2^2}{2nh_xh_y}\widehat{D}_1(\widehat{\boldsymbol{\beta}}) - \frac{\nu_2}{2nh_x}\widehat{D}_2(\widehat{\boldsymbol{\beta}}) - \frac{\nu_2}{2nh_y}\widehat{D}_3.$$

Apparently, the null hypothesis will be more likely to be rejected when T_n is larger. Based on Theorem 1, T_n converges to $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^*, Y)$ with probability tending to 1 when $\mathbf{x}$ and Y are independent. The next theorem derives the asymptotic normality of the testing statistics T_n under the null hypothesis.

Theorem 3. *Assume regularities conditions (C1)-(C4) hold. When $c_n nh_x^{1/2}h_y^{1/2}/\log n \to \infty$, under the null hypothesis,*

$$nh_x^{1/2}h_y^{1/2}T_n \xrightarrow{d} N\left[0, 2^{-1}\nu_2^2 E\left\{f^{-1}_{\mathbf{x}^T\boldsymbol{\beta}^*,Y}(\mathbf{x}^T\boldsymbol{\beta}^*, Y)\right\}\right],$$

where $\nu_2 \stackrel{\text{def}}{=} \int K^2(u)du$.

To implement the test more accurately, the asymptotic null distribution of T_n is approximated through random permutations. Specifically, we obtain replicates of the testing statistics under random permutations of Y, which is defined as $\widetilde{Y}$, while keeping $\mathbf{x}$ fixed. The p-value is calculated as the fraction of replicates of $\widetilde{T}_n = \widetilde{U}_n(\boldsymbol{\beta}^*) = \widehat{I}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^*, \widetilde{Y})$ that are at least as large as the observed test statistic. The validity of random permutation method follows because the sequence of test statistics is exchangeable under the null hypothesis. A detailed introduction of permutation tests could be found in Good (2006) and Pesarin and Salmaso (2010).

In what follows, we discuss the asymptotic properties under the fixed alternative and a sequence of local alternatives. Under the fixed alternative, the mutual information

is bounded away from zero, i.e.,

$$H_{1f} : I(\mathbf{x}, Y) \geq c > 0.$$

Note that

$$\begin{aligned} T_n &= U_n(\widehat{\boldsymbol{\beta}}) - c_n I(\widehat{\boldsymbol{\beta}} \neq \boldsymbol{\beta}^*) + c_n I(\widehat{\boldsymbol{\beta}} \neq \boldsymbol{\beta}^*) \\ &\geq \max_{\|\boldsymbol{\beta}\|=1} \{U_n(\boldsymbol{\beta}) - c_n \mathbb{I}(\boldsymbol{\beta} \neq \boldsymbol{\beta}^*)\} \\ &\geq U_n(\boldsymbol{\beta}) - c_n \mathbb{I}(\boldsymbol{\beta} \neq \boldsymbol{\beta}^*) = I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y) + o_p(1), \quad \forall\, \boldsymbol{\beta}. \end{aligned}$$

Under the fixed alternative hypothesis, there exists at least one $\boldsymbol{\beta}$ such that $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}$ and Y are dependent, i.e., $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y) \geq c' > 0$, indicating that the test is consistent. We prove the result formally in Theorem 4.

Theorem 4. *Assume the regularities conditions (C1)-(C4) hold. Under the fixed alternative hypothesis, when $c_n \to 0$,*

$$nh_x^{1/2} h_y^{1/2} T_n \to \infty.$$

By further assuming that $\boldsymbol{\beta}_0$ is unique and condition (C5) holds, as $c_n n^{1/2} \to 0$,

$$n^{1/2} \left\{T_n - I(\mathbf{x}^T\boldsymbol{\beta}_0, Y)\right\} \xrightarrow{d} N\left(0, \operatorname{var}\left[\log\left\{\frac{f_{Y|\mathbf{x}^T\boldsymbol{\beta}_0}(Y \mid \mathbf{x}^T\boldsymbol{\beta}_0)}{f_Y(Y)}\right\}\right]\right).$$

Besides consistency, Theorem 4 also states that when the maximum unit direction $\boldsymbol{\beta}_0$ is unique, T_n is a root-n consistent estimator of $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0, Y)$, which can be used to infer about $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0, Y)$. Next, we consider a sequence of local alternatives with the

form

$$H_{1l}: \quad f_{\mathbf{x},Y}(\mathbf{x},y) = f_{\mathbf{x}}(\mathbf{x})f_Y(y)\{1+\lambda_n g(\mathbf{x},y)\},$$

where the function $g(\cdot)$ satisfies 1) $1+\lambda_n g(\mathbf{x},y) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^p$ and $y \in \mathbb{R}$, and 2) $\int g(\mathbf{x},y)f_{\mathbf{x}}(\mathbf{x})f_Y(y)dy = \int g(\mathbf{x},y)f_{\mathbf{x}}(\mathbf{x})f_Y(y)d\mathbf{x} = 0$.

Under the local alternative, the mutual information between Y and $\mathbf{x}$ is of order $O(\lambda_n^2)$. In fact, simple calculations imply that $E\{g(\mathbf{x},Y)\} = O(\lambda_n)$, and $E\{g^2(\mathbf{x},Y)\} = O(1)$. As a result, we obtain that

$$\begin{aligned} & E\left\{\log \frac{f_{\mathbf{x},Y}(\mathbf{x},Y)}{f_{\mathbf{x}}(\mathbf{x})f_Y(Y)}\right\} = E\log\{1+\lambda_n g(\mathbf{x},Y)\} \\ = \; & \lambda_n E g(\mathbf{x},Y) - 2^{-1}\lambda_n^2 E g^2(\mathbf{x},Y) + o(\lambda_n^2) = O(\lambda_n^2). \end{aligned}$$

The asymptotic properties of T_n under the local alternative hypothesis is presented in Theorem 5.

Theorem 5. *Assume the regularities conditions (C1)-(C4) hold. Under the local alternative hypothesis, when* $nh_x^{1/2}h_y^{1/2}\lambda_n^2 \to \infty$, $\lambda_n^2/c_n \to \infty$,

$$nh_x^{1/2}h_y^{1/2}T_n \to \infty.$$

Recall that in Theorem 3, c_n should be large enough such that $c_n nh_x^{1/2}h_y^{1/2}/\log n \to \infty$ to control the size. Theorem 5 indicates that our test is powerful in detecting local alternatives as long as λ_n is larger than $n^{-1/2}h_x^{-1/4}h_y^{-1/4}\log n$ by taking $c_n = n^{-1}h_x^{-1/2}h_y^{-1/2}\log^2 n$. It implies that our proposed test can detect local alternatives converging to the null at the rate $o(n^{-1/4})$ under the mild conditions. Note that the local alternatives in Berrett and Samworth (2019) are defined in terms of the mutual

information, whose rate $b_n = o(n^{-1/2})$ is in fact equivalent to the rate $\lambda_n = o(n^{-1/4})$ in our paper. It should be further mentioned that the rates in Berrett and Samworth (2019) are minimax rates that hold uniformly over a wide class of alternatives whose mutual information is bounded below. Considering the fact that local smoothing based tests can generally achieve consistency against local alternatives at rate arbitrarily slower than $n^{-1/2}h_x^{-p/4}h_y^{-1/4}$, the rate in our paper is inherently faster than classical local smoothing based tests.

3. EXTENSIONS TO ARBITRARY DIMENSIONS

Most of the existing tests for independence developed recently have the capability of handling arbitrary dimensions (Zhu et al., 2017; Berrett and Samworth, 2019; Székely and Rizzo, 2013; Heller et al., 2012). In the previous sections we have stated how to tackle the situations where $\mathbf{x}$ has relative large dimension while $\mathbf{y}$ is relatively small dimensional. In this section, we discuss the scenario where both $\mathbf{x}$ and $\mathbf{y}$ have large dimensions. Assume that $\mathbf{x} \in \mathbb{R}^p$ and $\mathbf{y} \in \mathbb{R}^q$. Recall that the mutual information can characterize arbitrary dependence between $\mathbf{x}$ and $\mathbf{y}$, is non-negative, and equals zero if and only if $\mathbf{x} \perp\!\!\!\perp \mathbf{y}$. In addition, $\mathbf{x} \perp\!\!\!\perp \mathbf{y}$ holds if and only if $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta} \perp\!\!\!\perp \mathbf{y}^{\mathrm{T}}\boldsymbol{\alpha}$ for any $\boldsymbol{\alpha} \in \mathbb{R}^q$ and $\boldsymbol{\beta} \in \mathbb{R}^p$ such that $\|\boldsymbol{\alpha}\| = \|\boldsymbol{\beta}\| = 1$. That is, $I(\mathbf{x}, \mathbf{y}) \geq 0$, and the equality holds if and only if $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y}^{\mathrm{T}}\boldsymbol{\alpha}) = 0$ for all $\|\boldsymbol{\alpha}\| = \|\boldsymbol{\beta}\| = 1$. Furthermore, when $I(\mathbf{x}, \mathbf{y}) > 0$, i.e., $\mathbf{x} \not\perp\!\!\!\perp \mathbf{y}$, there must exists some $\boldsymbol{\alpha} \in \mathbb{R}^q$ and $\boldsymbol{\beta} \in \mathbb{R}^p$, such that $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y}^{\mathrm{T}}\boldsymbol{\alpha}) > 0$, where $\|\boldsymbol{\alpha}\| = \|\boldsymbol{\beta}\| = 1$. Therefore, to avoid the large dimensionality issue of either $\mathbf{x}$ or $\mathbf{y}$, we perform projections for both $\mathbf{x}$ and $\mathbf{y}$, and consider estimating $\sup_{\|\boldsymbol{\alpha}\|=\|\boldsymbol{\beta}\|=1} I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y}^{\mathrm{T}}\boldsymbol{\alpha})$, i.e., the maximum mutual information between projected $\mathbf{x}$ and $\mathbf{y}$. However, as discussed previously, when $\mathbf{x}$ is independent of $\mathbf{y}$, the maximum unit direction for both $\mathbf{x}$ and $\mathbf{y}$ are not identifiable. Hence, we also propose adding a penalty term $-c_n I\left\{(\boldsymbol{\alpha}, \boldsymbol{\beta}) \neq (\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*)\right\}$ to guarantee that the resultant estimator goes

to $(\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*)$ under the null and goes to the maximum unit direction under the alternative hypothesis. Specifically, define

$$(\widehat{\boldsymbol{\alpha}}, \widehat{\boldsymbol{\beta}}) = \underset{\|\boldsymbol{\alpha}\|=1, \|\boldsymbol{\beta}\|=1}{\arg\max} \left[V_n(\boldsymbol{\alpha}, \boldsymbol{\beta}) - c_n I \left\{ (\boldsymbol{\alpha}, \boldsymbol{\beta}) \neq (\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*) \right\} \right], \tag{3.1}$$

where $\boldsymbol{\alpha}^* \in \mathbb{R}^q$ and $\|\boldsymbol{\alpha}^*\| = 1$. $V_n(\boldsymbol{\alpha}, \boldsymbol{\beta})$ is an unbiased estimator of $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y}^{\mathrm{T}}\boldsymbol{\alpha})$, which is defined as

$$V_n(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^{n} \log \left\{ \frac{\widehat{f}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y}^{\mathrm{T}}\boldsymbol{\alpha}}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y}_i^{\mathrm{T}}\boldsymbol{\alpha})}{\widehat{f}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}) \widehat{f}_{\mathbf{y}^{\mathrm{T}}\boldsymbol{\alpha}}(\mathbf{y}_i^{\mathrm{T}}\boldsymbol{\alpha})} \right\} + \frac{\nu_2^2 \sum_{i=1}^{n} \widehat{f}^{-1}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y}^{\mathrm{T}}\boldsymbol{\alpha}}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, \mathbf{y}_i^{\mathrm{T}}\boldsymbol{\alpha})}{2n^2 h_x h_y}$$
$$- \frac{\nu_2 \sum_{i=1}^{n} \widehat{f}^{-1}_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{2n^2 h_x} - \frac{\nu_2 \sum_{i=1}^{n} \widehat{f}^{-1}_{\mathbf{y}^{\mathrm{T}}\boldsymbol{\alpha}}(\mathbf{y}_i^{\mathrm{T}}\boldsymbol{\alpha})}{2n^2 h_y}.$$

To solve for the $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, we apply Algorithm 1 repeatedly. The details are illustrated in Algorithm 2 below.

Algorithm 2 Estimating the maximum unit direction under large dimensional settings

1. Input error control ϵ, maximum iteration S, and initial values $\boldsymbol{\alpha}_0$ and $\boldsymbol{\beta}_0$.
2. While $1 \leq s \leq S$
   (1) Apply Step 2 in Algorithm 1 to the data $\{\mathbf{x}_i, \mathbf{y}_i^{\mathrm{T}}\boldsymbol{\alpha}_{s-1}\}$, $i = 1, \ldots, n$ to obtain $\boldsymbol{\beta}_s$.
   (2) Apply Step 2 in Algorithm 1 to the data $\{\mathbf{y}_i, \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_s\}$, $i = 1, \ldots, n$ to obtain $\boldsymbol{\alpha}_s$.
   (3) Break if $\|\boldsymbol{\beta}_s - \boldsymbol{\beta}_{s-1}\|_2 + \|\boldsymbol{\alpha}_s - \boldsymbol{\alpha}_{s-1}\|_2 \leq \epsilon$.
3. Output $(\widehat{\boldsymbol{\alpha}}, \widehat{\boldsymbol{\beta}}) = (\boldsymbol{\alpha}_s, \boldsymbol{\beta}_s)$ if $V_n(\boldsymbol{\alpha}_s, \boldsymbol{\beta}_s) - c_n I \left\{ (\boldsymbol{\alpha}_s, \boldsymbol{\beta}_s) \neq (\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*) \right\} > V_n(\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*)$ and $(\widehat{\boldsymbol{\alpha}}, \widehat{\boldsymbol{\beta}}) = (\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*)$ otherwise.

4. NUMERICAL ANALYSIS

To implement our proposed test, we first discuss how to choose $\boldsymbol{\beta}^*$. Our theoretical conclusions hold for an arbitrary $\boldsymbol{\beta}^*$. But $\boldsymbol{\beta}^*$ still matters from a practical perspective. We suggest choosing $\boldsymbol{\beta}^* = (1, 1, \ldots, 1)^{\mathrm{T}}/p^{1/2}$ where there is no prior information about

the alternative. If one suspects that some coordinates of $\mathbf{x}$ deviate more substantially from the null, one may set the corresponding components of $\boldsymbol{\beta}^*$ to be relatively larger than others. In our test, c_n should be relatively large, i.e., $\left(c_n n h_x^{1/2} h_y^{1/2} / \log n\right) \to \infty$, to control the size. And $c_n \to 0$ is required to avoid power loss under the alternatives. Thus in practice, after fixing $\boldsymbol{\beta}^*$, we choose c_n as follows. Let $\widetilde{Y}$ be a permuted sample of Y. Based on $(\mathbf{x}, \widetilde{Y})$, we define

$$d_n \stackrel{\text{def}}{=} \sup_{\|\boldsymbol{\beta}\|=1} \widehat{I}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, \widetilde{Y}) - \widehat{I}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^*, \widetilde{Y}).$$

We repeat this random permutation procedure 1000 times and choose c_n to be the 0.99 quantile of d_n. This method works very well numerically, as shown in the following subsections. It could ensure that the maximum unit direction converges to $\boldsymbol{\beta}^*$ under the null, while avoiding unnecessary power loss under the alternative.

4.1. Simulation Study

In this section, we perform extensive numerical analysis to illustrate the effectiveness of the proposed independence test. Specifically, we conduct simulations to compare the proposed test with another mutual information test proposed by Berrett and Samworth (2019), distance correlation (Székely and Rizzo, 2013) and ranks of distances (Heller et al., 2012). All three competing tests are implemented in R. The test based on mutual information is implemented by calling the "MINTperm" function from the "IndepTest" package. The test based on distance correlation is implemented by calling the "dcor.ttest" function from the "energy" package while the the test based on ranks of distances is built in the "hhg.test" function from the "HHG" package. We use 1000 permutations in each settings to estimate the p-value and repeat each setting 1000 times. The size and power of the tests are reported at two significance levels, i.e., $\alpha = 0.05$ and 0.1. Consider the following four scenarios, where $\mathbf{x} = (X_1, \ldots, X_p)^{\mathrm{T}} \sim$

$N(0, \mathbf{I}_p)$, $\varepsilon \sim N(0,1)$ is an independent error term and a increases from 0 to 1. We also set $\boldsymbol{\beta}_1 = (1,1,0,\ldots,0)^{\mathrm{T}}$ and $\boldsymbol{\beta}_2 = (1,0,\ldots,0)^{\mathrm{T}}$ with dimension $p = 10$. In each scenario, the sample size varies from 50 to 100, respectively.

$$
\begin{aligned}
&\text{Scenario 1.} \qquad Y = a(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_1) + \varepsilon;\\
&\text{Scenario 2.} \qquad Y = a(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_1)^2 + \varepsilon;\\
&\text{Scenario 3.} \qquad Y = a(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_1)(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_2) + 0.5\varepsilon;\\
&\text{Scenario 4.} \qquad Y = a\sum_{k=1}^{5}\{\sin(X_{2k-1}) + \cos(X_{2k})\} + \varepsilon.
\end{aligned}
$$

In all scenarios, the intensity parameter a controls whether each model is under the null or the alternative hypothesis. When $a = 0$, Y is independent of $\mathbf{x}$. It corresponds to the null hypothesis. When $a > 0$, in Scenario 1, Y is linearly dependent of $\mathbf{x}$. For Scenario 2, there is a nonlinear and nonmonotone dependence between Y and $\mathbf{x}$. As for Scenario 3, Y depends on $\mathbf{x}$ through two different linear combinations. For the last Scenario, Y is additively related to all covariates of $\mathbf{x}$ and any linear combination of $\mathbf{x}$ is not sufficient to characterize the relationship between $\mathbf{x}$ and Y.

We first use Scenario 1 to study the sensitivity of our proposed test to the tuning parameters, particularly to the pre-specified direction $\boldsymbol{\beta}^*$. For the kernel bandwidths, we use rule of thumb in our simulation study for simplicity. We set $\boldsymbol{\beta}^* = (1,1,\ldots,1)^{\mathrm{T}}/10^{1/2}$ if we do not have any prior information about the alternative. This direction is simple because we put the same weights for all covariates. As a comparison, we set $\boldsymbol{\beta}^* = \boldsymbol{\beta}_1/2^{1/2}$, which is proportional to the maximum unit direction under the alternative. This means that we believe Y is either independent of $\mathbf{x}$ or most likely related to $\mathbf{x}$ through $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_1$. In this sensitivity study, we only report the empirical sizes and powers for Scenario 1 at significance level $\alpha = 0.05$ when the sample size $n = 50$

in Table 1. It can be clearly seen that, the test based on $\boldsymbol{\beta}^* = (1, 1, 0, \ldots, 0)^{\mathrm{T}}/2^{1/2}$ is slightly more powerful than that of $\boldsymbol{\beta}^* = (1, 1, \ldots, 1)^{\mathrm{T}}/10^{1/2}$. This meets our expectation.

Table 1: Empirical size and power of the proposed test for different choices of $\boldsymbol{\beta}^*$ in Scenario 1.

$\boldsymbol{\beta}^*$	$a = 0$	$a = 0.2$	$a = 0.4$	$a = 0.6$	$a = 0.8$	$a = 1$
$(1, 1, \ldots, 1)^{\mathrm{T}}/10^{1/2}$	0.058	0.074	0.216	0.461	0.794	0.962
$(1, 1, 0 \ldots, 0)^{\mathrm{T}}/2^{1/2}$	0.054	0.279	0.773	0.978	1.000	1.000

Next, we fix $\boldsymbol{\beta}^* = (1, 1, \ldots, 1)^{\mathrm{T}}/p^{1/2}$ because we have no knowledge about the favored alternative. The simulation results of the empirical sizes and powers for all scenarios are summarized in Tables 2 to 6, respectively. It can be clearly seen that, the empirical type-I error rates of all tests are close to the nominal significance levels. In addition, all these tests are consistent in a sense that, when $a > 0$, the empirical power goes to 1 in each scenario as the sample size increases. Among all the testing procedures, our proposed method is the most powerful one in most scenarios, especially when the sample size is small. For example, consider scenario 2 when sample size is 50, $a = 0.8$. According to Table 3, the empirical power of the proposed method is 0.848 when the significance level is 0.05, while the other three tests' powers are smaller than 0.5. That is, the proposed test also exhibits good capability for testing nonlinear and nonmonotone dependence like scenario 2. Although the proposed method only seeks for the maximum unit direction, from Table 4, we can see that it still outperforms the other methods even when one direction may not be sufficient to characterize the relationship between Y and $\mathbf{x}$. In addition, as we can see from Table 5, for a more general additive model rather than index structured model, our test is still comparable with other methods. Overall, these results indicate that the proposed test is very powerful in detecting dependence when the size is controlled at a reasonable level.

Table 2: Empirical size and power of the proposed test (MI), mutual information estimated by entropy (BS), distance correlation (DC), and ranks of distances (HG) for different a in Scenario 1. Sample sizes vary from 50 to 100 among the sub scenarios.

n	α	Test	$a = 0$	$a = 0.2$	$a = 0.4$	$a = 0.6$	$a = 0.8$	$a = 1$
50	0.05	MI	0.058	0.074	0.216	0.461	0.794	0.962
		BS	0.047	0.079	0.218	0.482	0.645	0.785
		DC	0.065	0.198	0.608	0.910	0.988	0.996
		HG	0.042	0.079	0.176	0.467	0.648	0.810
	0.1	MI	0.101	0.138	0.299	0.550	0.837	0.974
		BS	0.098	0.153	0.330	0.620	0.757	0.876
		DC	0.110	0.267	0.708	0.946	0.995	0.999
		HG	0.085	0.146	0.295	0.601	0.751	0.881
100	0.05	MI	0.064	0.117	0.430	0.874	0.990	1.000
		BS	0.046	0.101	0.501	0.865	0.978	0.998
		DC	0.062	0.393	0.947	1.000	1.000	1.000
		HG	0.040	0.095	0.471	0.832	0.961	0.993
	0.1	MI	0.112	0.185	0.526	0.914	0.995	1.000
		BS	0.111	0.209	0.644	0.935	0.989	0.998
		DC	0.100	0.468	0.970	1.000	1.000	1.000
		HG	0.085	0.182	0.607	0.908	0.981	0.995

Next, we investigate the dimension effect for both $\mathbf{x}$ and $\mathbf{y}$. Suppose we draw $\mathbf{x} = (X_1, \ldots, X_p)^{\mathrm{T}}$ from a multivariate normal distribution with mean zero and covariance matrix $\mathbf{I}_p$, the error term $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_q)^{\mathrm{T}} \sim N(0, \mathbf{I}_q)$. We generate $\mathbf{y} = (Y_1, \ldots, Y_q)^{\mathrm{T}}$ as follows. Let $Y_1 = a\cos(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_1) + 0.5\exp(-a|\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_1|)\varepsilon_1$, and $Y_j = 0.5\varepsilon_j$, for $j = 2, \ldots, q$, where $\boldsymbol{\beta}_1 = (1, 1, 0, \ldots, 0)^{\mathrm{T}}$. We let p and q vary among 2, 5 and 10 and study the performances of our test under both null and alternative hypothesises by letting $a = 0$ and $a = 1$, respectively. The empirical size and power performances are shown in Table 6. Since we estimate the maximum unit projection for both $\mathbf{x}$ and $\mathbf{y}$ in this case and the resultant test statistic behaves almost as if the model was univariate, we can clearly see that the power performance of our method is competitive with respect to other existing tests, especially when p or q increases. For example, the power of our test when $a = 1$, $p = q = 10$ is 0.815 at the significance level 0.05, while the empirical power of other tests are all smaller than 0.2.

Table 3: Empirical size and power of the proposed test (MI), mutual information by entropy (BS), distance correlation (DC), and ranks of distances (HG) for different a in Scenario 2..

n	α	Test	$a = 0$	$a = 0.2$	$a = 0.4$	$a = 0.6$	$a = 0.8$	$a = 1$
50	0.05	MI	0.055	0.139	0.286	0.548	0.701	0.819
		BS	0.047	0.128	0.260	0.368	0.438	0.465
		DC	0.065	0.086	0.126	0.136	0.154	0.155
		HG	0.042	0.137	0.215	0.316	0.328	0.349
	0.1	MI	0.098	0.232	0.384	0.629	0.783	0.867
		BS	0.098	0.240	0.388	0.541	0.576	0.619
		DC	0.110	0.144	0.179	0.206	0.214	0.221
		HG	0.085	0.205	0.311	0.429	0.444	0.479
100	0.05	MI	0.059	0.241	0.813	0.976	0.999	1.000
		BS	0.046	0.198	0.486	0.643	0.718	0.770
		DC	0.062	0.094	0.157	0.207	0.254	0.269
		HG	0.040	0.175	0.417	0.559	0.640	0.668
	0.1	MI	0.099	0.339	0.853	0.986	0.999	1.000
		BS	0.111	0.313	0.643	0.787	0.830	0.873
		DC	0.100	0.148	0.224	0.307	0.359	0.353
		HG	0.085	0.268	0.546	0.685	0.737	0.773

4.2. A Real Data Example

In this section, we apply the proposed test to a stock portfolio performance data set (Liu and Yeh, 2017) from the UCI machine learning repository. The data contains 63 observations of quarterly recorded financial data which contains six predictors: the weight of large B/P concept, the weight of large ROE concept, the weight of large S/P concept, the weight of large return rate in the last quarter concept, the weight of large market value concept, and the weight of small systematic risk concept. We examine whether the absolute winning rate are independent of the six predictors. We also compare our proposed test with BS (Berrett and Samworth, 2019), DC (Székely and Rizzo, 2013) and HG (Heller et al., 2012). The p-values of these tests are given below in Table 7. It can be seen that all the p-values are smaller than 0.05, indicating that the absolute winning rate is indeed dependent of the six predictors, at significance level 0.05. To see this, we chart scatterplots of the absolute winning rate versus the standardized

Table 4: Empirical size and power of the proposed test (MI), mutual information estimated by entropy (BS), distance correlation (DC), and ranks of distances (HG) for different a in Scenario 3..

n	α	Test	$a = 0$	$a = 0.2$	$a = 0.4$	$a = 0.6$	$a = 0.8$	$a = 1$
50	0.05	MI	0.051	0.155	0.486	0.694	0.904	0.922
		BS	0.039	0.152	0.311	0.419	0.454	0.513
		DC	0.065	0.086	0.117	0.148	0.155	0.165
		HG	0.042	0.125	0.227	0.321	0.357	0.369
	0.1	MI	0.094	0.227	0.547	0.750	0.936	0.956
		BS	0.094	0.232	0.441	0.569	0.633	0.662
		DC	0.110	0.149	0.180	0.218	0.219	0.235
		HG	0.085	0.208	0.339	0.432	0.461	0.479
100	0.05	MI	0.055	0.329	0.963	0.998	1.000	1.000
		BS	0.051	0.289	0.668	0.815	0.861	0.860
		DC	0.062	0.095	0.179	0.209	0.253	0.270
		HG	0.040	0.220	0.442	0.582	0.656	0.688
	0.1	MI	0.113	0.401	0.972	0.999	1.000	1.000
		BS	0.102	0.433	0.791	0.905	0.932	0.926
		DC	0.100	0.159	0.266	0.302	0.341	0.363
		HG	0.085	0.307	0.573	0.694	0.771	0.793

$(\mathbf{x}_1^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_1)$ in Panels (A) of Figure 1, where $\mathbf{x}_1$ is the vector of the six predictors and $\widehat{\boldsymbol{\beta}}_1$ is obtained through maximizing (2.1). The solid lines are the fitted values of local linear regression, with the bandwidths decided by leave-one-out cross-validation. The dashed lines are the 95% pointwise confidence intervals. Accordingly, we can clearly see the nonlinear dependence between the absolute winning rate and the predictors.

To make the problem more challenging, we add other six predictors whose entries are randomly and independently generated from $N(0, 1)$. We also summarize the corresponding p-values of these tests in Table 7. It can be seen that at significance level 0.05, our proposed test is still powerful at detecting the dependence between the absolute winning rate and the predictors, even the dimension increases to 12. However, all other three tests failed to reject the null hypothesis, which confirms that these methods are sensitive to the dimension. We also chart the corresponding scatterplots of the absolute winning rate versus the standardized $(\mathbf{x}_2^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_2)$ in Panels (B) of Figure 1, which

Table 5: Empirical size and power of the proposed test (MI), mutual information estimated by entropy (BS), distance correlation (DC), and ranks of distances (HG) for different a in Scenario 4.

n	α	Test	$a = 0$	$a = 0.2$	$a = 0.4$	$a = 0.6$	$a = 0.8$	$a = 1$
50	0.05	MI	0.063	0.116	0.319	0.513	0.681	0.817
		BS	0.061	0.084	0.224	0.364	0.460	0.553
		DC	0.076	0.173	0.599	0.825	0.936	0.968
		HG	0.050	0.104	0.374	0.586	0.753	0.848
	0.1	MI	0.111	0.202	0.429	0.641	0.779	0.880
		BS	0.110	0.154	0.323	0.484	0.594	0.687
		DC	0.115	0.259	0.697	0.881	0.964	0.984
		HG	0.110	0.182	0.497	0.706	0.859	0.917
100	0.05	MI	0.071	0.174	0.537	0.820	0.945	0.979
		BS	0.058	0.156	0.487	0.820	0.948	0.976
		DC	0.072	0.382	0.940	0.994	1.000	1.000
		HG	0.048	0.222	0.703	0.945	0.987	0.998
	0.1	MI	0.121	0.271	0.658	0.880	0.967	0.995
		BS	0.121	0.244	0.620	0.886	0.975	0.992
		DC	0.116	0.474	0.969	0.999	1.000	1.000
		HG	0.098	0.334	0.809	0.971	0.994	0.999

shows similar nonlinear dependence patten between the absolute winning rate and the predictors.

5. Discussion

In this paper, we develop a new nonparametric independence test based on projected mutual information, where the maximum unit direction is estimated via penalized mutual information. A fast algorithm is also introduced to estimate the maximum unit direction. The proposed test is shown to be consistent against all dependent alternatives and has asymptotic power one for local alternatives at rate arbitrarily slower than $n^{-1/2}h_x^{-1/4}h_y^{-1/4}/\log n$. Numerical results indicate that the test is more powerful compared with other traditional testing procedures, especially when the dimension size is large or sample size is small.

The theoretical and numerical results in this paper are very encouraging. Con-

Table 6: Empirical size and power of the proposed test (MI), mutual information estimated by entropy (BS), distance correlation (DC), and ranks of distances (HG) for different dimension size. The sample size n is 50. $a = 0$ stands for the case where the null hypothesis is true, while $a = 1$ stands for the case where the alternative hypothesis is true.

a	α	Test	$p = 2$			$p = 5$			$p = 10$		
			$q = 2$	$q = 5$	$q = 10$	$q = 2$	$q = 5$	$q = 10$	$q = 2$	$q = 5$	$q = 10$
0	0.05	MI	0.057	0.054	0.063	0.054	0.062	0.061	0.064	0.050	0.060
		BS	0.040	0.045	0.046	0.043	0.045	0.060	0.054	0.043	0.058
		DC	0.062	0.080	0.063	0.065	0.060	0.056	0.056	0.069	0.067
		HG	0.043	0.054	0.042	0.049	0.043	0.051	0.052	0.045	0.057
	0.1	MI	0.095	0.106	0.104	0.099	0.112	0.108	0.124	0.102	0.113
		BS	0.093	0.102	0.102	0.096	0.095	0.102	0.112	0.089	0.110
		DC	0.098	0.124	0.112	0.099	0.105	0.104	0.096	0.111	0.113
		HG	0.086	0.093	0.086	0.087	0.097	0.103	0.093	0.100	0.105
1	0.05	MI	1.000	0.998	0.966	0.998	0.991	0.901	0.982	0.881	0.815
		BS	0.730	0.467	0.305	0.288	0.201	0.130	0.144	0.105	0.099
		DC	0.894	0.755	0.621	0.281	0.215	0.171	0.126	0.129	0.104
		HG	0.997	0.938	0.827	0.614	0.490	0.349	0.319	0.237	0.169
	0.1	MI	1.000	1.000	0.981	0.998	0.996	0.920	0.986	0.904	0.845
		BS	0.814	0.607	0.446	0.427	0.309	0.217	0.246	0.196	0.163
		DC	0.942	0.843	0.718	0.372	0.307	0.260	0.199	0.183	0.171
		HG	0.999	0.976	0.888	0.717	0.593	0.445	0.423	0.342	0.263

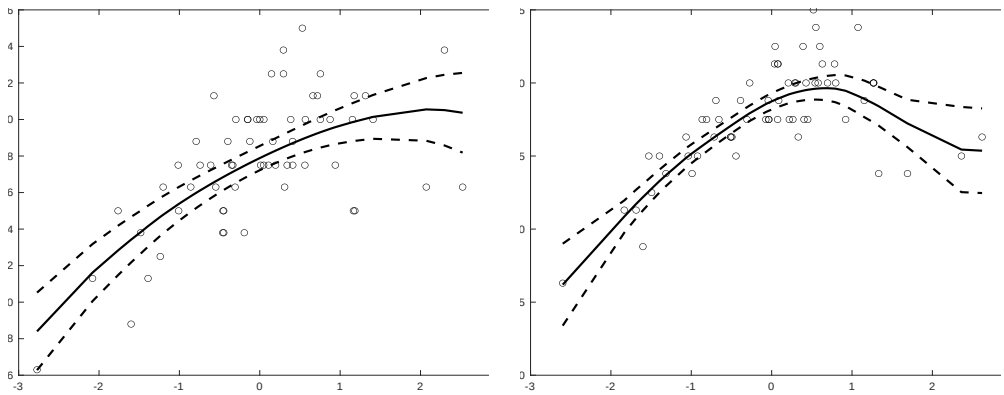
Table 7: p-value of the proposed test together with mutual information estimated by entropy (BS), distance correlation (DC), and ranks of distances (HG)

Number of predictors	Test	MI	BS	DC	HG
6	p-value	<0.001	0.014	<0.001	0.003
12	p-value	<0.001	0.229	0.441	0.389

ditional independence plays a very important role in many statistical and economic problems. See for instance Su and White (2014); Huang et al. (2016) and Wang and Hong (2018). It would be interesting to extend the basic idea introduced in this paper to test conditional independence. We leave it for future work.

APPENDIX: PROOF OF LEMMAS AND THEOREMS

Appendix A1: Proof of Lemma 1



(A) (B)

Figure 1: The scatterplots of the absolute winning rate versus standardized $(\mathbf{x}_1^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_1)$ in Panel (A) and versus standardized $(\mathbf{x}_2^{\mathrm{T}}\widehat{\boldsymbol{\beta}}_2)$ in Panel (B) in the Real Data Example.

First, according to Proposition 4.3 of Cook (2009), Y is independent of $\mathbf{x}$ if and only if $Y \perp\!\!\!\perp \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}$ for all $\boldsymbol{\beta} \in \mathbb{R}^p$. Meanwhile, because mutual information between $\mathbf{y}$ and $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}$ is nonnegative and equals zero if and only if $\mathbf{y}$ and $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}$ are independent, $Y \perp\!\!\!\perp \mathbf{x}$ is further equivalent to $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y) = 0$ for all $\boldsymbol{\beta} \in \mathbb{R}^p$, which holds true if and only if $\sup_{\|\boldsymbol{\beta}\|=1} I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y) = 0$. Therefore, the proof is completed. □

Appendix A2: Proof of Theorem 1

For convenience, we neglect the subscript for the density functions $f_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\cdot)$, $f_Y(\cdot)$, $f_{\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta},Y}(\cdot)$ and $f_{Y|\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}}(\cdot)$ in our proof. We first show that

$$\begin{aligned} n^{-1}\sum_{i=1}^{n}\log\left\{\frac{\widehat{f}(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{\widehat{f}(Y_i)}\right\} &= n^{-1}\sum_{i=1}^{n}\log\left\{\frac{f(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(Y_i)}\right\} + 1 + n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)} \\ &\quad -n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})} - n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(Y_i)}{f(Y_i)} + R_1 + R_2 - R_3 + o_p(n^{-1}h_x^{-1/2}h_y^{-1/2}), \quad \text{(A2.1)} \end{aligned}$$

holds uniformly for all $\boldsymbol{\beta}$, and R_1, R_2 and R_3 are defined as

$$R_1 = n^{-1}\sum_{i=1}^{n}\frac{\left\{\widehat{f}(Y_i) - f(Y_i)\right\}^2}{2f^2(Y_i)}, \quad R_2 = n^{-1}\sum_{i=1}^{n}\frac{\left\{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}) - f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})\right\}^2}{2f^2(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})},$$

$$R_3 = n^{-1} \sum_{i=1}^{n} \frac{\left\{ \widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i) - f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i) \right\}^2}{2f^2(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)}.$$

To see this, it suffices to verify

$$\log \widehat{f}(\mathbf{u}) - \log f(\mathbf{u}) - \frac{\widehat{f}(\mathbf{u})}{f(\mathbf{u})} + 1 - \frac{\left\{ \widehat{f}(\mathbf{u}) - f(\mathbf{u}) \right\}^2}{2f^2(\mathbf{u})} \tag{A2.2}$$

is of order $o_p(n^{-1}h_x^{-1/2}h_y^{-1/2})$ and holds uniformly for all $\mathbf{u}$. Here $\mathbf{u}$ can be y, $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}$ or $(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, y)$. We first show it holds when $\mathbf{u}$ is chosen to be y. To begin with,

$$\begin{aligned} \sup_y \left| E\widehat{f}(y) - f(y) \right| &= \sup_y \left| \int \frac{1}{h_y} K\left(\frac{y-u}{h_y} \right) \{f(u) - f(y)\} \, du \right| \\ &= \sup_y \left| \int K(v) \{f(y - vh_y) - f(y)\} \, dv \right|, \end{aligned}$$

where we let $v = (y-u)/h_y$. According to condition (C1), following Lemma 3.3 of Zhu and Fang (1996), by expanding $f(y - vh_y)$ with the Lagrange form of the remainder term, it equals

$$\begin{aligned} & \sup_y \left| \int K(v) f^{(m-1)}(y - \delta v h_y)(vh_y)^{m-1} dv \right| / (m-1)! \\ = & \sup_y \left| \int K(v) \left\{ f^{(m-1)}(y - \delta v h_y) - f^{(m-1)}(y) \right\} (vh_y)^{m-1} dv \right| / (m-1)!, \end{aligned}$$

where $0 < \delta < 1$ and the last equality holds because $\int v^{m-1} K(v) dv = 0$. Because the (m-1)-th derivative of $f(y)$ is locally Lipschitz-continuous, we have $f^{(m-1)}(y - \delta v h_y) - f^{(m-1)}(y)$ is bounded by $O(h_y)$. Thus we have $\sup_y \left| E\widehat{f}(y) - f(y) \right| = O(h_y^m)$. Meanwhile, according to Theorem 4 of Hansen (2008), we have $\sup_y \left| \widehat{f}(y) - E\widehat{f}(y) \right|$ is

of order $O_p(n^{-1/2}h_y^{-1/2}\log n)$. Combining these results, we have

$$\sup_y \left|\widehat{f}(y) - f(y)\right| \;=\; O_p(h_y^m + n^{-1/2}h_y^{-1/2}\log n). \tag{A2.3}$$

With Taylor's expansion, and by noting that $f(y)$ is bounded away from zero, when $nh_x^{1/2}h_y^{3m+1/2} \to 0$, and $nh_xh_y^2 \to \infty$, we have $\log \widehat{f}(y) = \log f(y) + \frac{\widehat{f}(y)}{f(y)} - 1 + \frac{\{\widehat{f}(y)-f(y)\}^2}{2f^2(y)} + o_p(n^{-1}h_x^{-1/2}h_y^{-1/2})$, where the remainder holds uniformly for all y. That is, (A2.2) holds when $\mathbf{u}$ is chosen to be y. Similarly, when $nh_x^{3m+1/2}h_y^{1/2} \to 0$, $nh_x^{1/2}h_y^{3m+1/2} \to 0$, and $nh_x^2h_y^2 \to \infty$, we have (A2.2) holds when $\mathbf{u}$ is chosen to be either $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}$ or $(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, y)$. Therefore, (A2.1) holds true.

Next, we simplify (A2.1). We first deal with R_1, where

$$\begin{aligned} R_1 &= n^{-3}\sum_{j\neq i, k\neq i} \frac{\left\{K_{h_y}(Y_j - Y_i) - f(Y_i)\right\}\left\{K_{h_y}(Y_k - Y_i) - f(Y_i)\right\}}{2f^2(Y_i)} \\ &= n^{-3}\sum_{i\neq j} \frac{\left\{K_{h_y}(Y_j - Y_i) - f(Y_i)\right\}^2}{2f^2(Y_i)} \\ &\quad + n^{-3}\sum_{i\neq j\neq k} \frac{\left\{K_{h_y}(Y_j - Y_i) - f(Y_i)\right\}\left\{K_{h_y}(Y_k - Y_i) - f(Y_i)\right\}}{2f^2(Y_i)}. \end{aligned} \tag{A2.4}$$

The first term is simplified as

$$\begin{aligned} n^{-3}\sum_{i\neq j}^{n} \frac{\left\{K_{h_y}(Y_j - Y_i) - f(Y_i)\right\}^2}{2f^2(Y_i)} &= n^{-3}\sum_{i\neq j} \frac{K_{h_y}^2(Y_j - Y_i)}{2f^2(Y_i)} + o_p(n^{-1}h_y^{-1/2}) \\ &= \frac{\nu_2}{2n^2h_y}\sum_{i=1}^{n} f^{-1}(Y_i) + o_p(n^{-1}h_y^{-1/2}), \end{aligned}$$

where $\nu_2 \stackrel{\text{def}}{=} \int K^2(u)du$. The first equality holds because $n^{-1}\sum_{j=1}^n K_{h_y}(Y_j - Y_i)$ equals $f(Y_i) + o_p(h_y^{-1/2})$ and the second equality holds because $n^{-1}\sum_{j=1}^n K_{h_y}^2(Y_j - Y_i)$ equals $E\{K_{h_y}^2(Y - Y_i) \mid Y_i\} + o_p(h_y^{-1/2})$ and $E\{K_{h_y}^2(Y - Y_i) \mid Y_i\} = h_y^{-1}\nu_2 f(Y_i) + o_p(h_y^{-1/2})$.

Now we deal with the second term of (A2.4). When $i \neq j \neq k$, because

$$E\left\{\frac{K_{h_y}(Y_j - Y_i)K_{h_y}(Y_k - Y_i)}{f^2(Y_i)}\middle|(Y_j, Y_k)\right\} = h_y^{-1}\int K(u)K\left(\frac{uh_y + Y_j - Y_k}{h_y}\right)f^{-1}(uh_y + Y_j)du,$$

by expanding $K\left(\frac{uh_y+Y_j-Y_k}{h_y}\right)f^{-1}(uh_y + Y_j)$ with Taylor series, we have

$$\begin{aligned}
&\frac{1}{n^2}\sum_{j\neq k}E\left[\frac{\left\{K_{h_y}(Y_j - Y_i) - f(Y_i)\right\}\left\{K_{h_y}(Y_k - Y_i) - f(Y_i)\right\}}{2f^2(Y_i)}\middle|(Y_j, Y_k)\right] \\
&-\frac{1}{n^2}\sum_{j\neq k}\left\{\frac{K_{h_y}(Y_j - Y_k) - f(Y_j)}{4f(Y_j)} + \frac{K_{h_y}(Y_j - Y_k) - f(Y_j)}{4f(Y_k)}\right\} \\
= \;& h_y^m\int u^m K(u)du\frac{1}{n}\sum_{k=1}^{n}\frac{\partial^m}{\partial Y_k^m}\left\{\frac{n^{-1}\sum_{j\neq k}K_h(Y_k - Y_j)}{2f(Y_k)}\right\}\{1 + o_p(1)\} \\
= \;& h_y^m\int u^m K(u)du\frac{1}{n}\sum_{k=1}^{n}\frac{\partial^m}{\partial Y_k^m}\left\{\frac{f(Y_k) + O_p(h_y^m + n^{-1/2}h_y^{-1/2})}{2f(Y_k)}\right\}\{1 + o_p(1)\} \\
= \;& O_p(h_y^{2m} + n^{-1/2}h_y^{m-1/2}).
\end{aligned}$$

That is,

$$\begin{aligned}
&\frac{1}{n^2}\sum_{j\neq k}E\left[\frac{\left\{K_{h_y}(Y_j - Y_i) - f(Y_i)\right\}\left\{K_{h_y}(Y_k - Y_i) - f(Y_i)\right\}}{2f^2(Y_i)}\middle|(Y_j, Y_k)\right] \\
= \;& \frac{1}{n^2}\sum_{j\neq k}\frac{K_{h_y}(Y_j - Y_k) - f(Y_j)}{2f(Y_j)} + O_p(h_y^{2m} + n^{-1/2}h_y^{m-1/2}).
\end{aligned}$$

Similarly, we can derive that

$$\begin{aligned}
&\frac{1}{n^2}\sum_{i\neq j}E\left[\frac{\left\{K_{h_y}(Y_j - Y_i) - f(Y_i)\right\}\left\{K_{h_y}(Y_k - Y_i) - f(Y_i)\right\}}{2f^2(Y_i)}\middle|(Y_i, Y_j)\right], \\
&\frac{1}{n^2}\sum_{i\neq k}E\left[\frac{\left\{K_{h_y}(Y_j - Y_i) - f(Y_i)\right\}\left\{K_{h_y}(Y_k - Y_i) - f(Y_i)\right\}}{2f^2(Y_i)}\middle|(Y_i, Y_k)\right]
\end{aligned}$$

are both of order $O_p(h_y^{2m} + n^{-1/2}h_y^{m-1/2})$. As a consequence, we conclude that

$$R_1 = \frac{\nu_2}{2n^2 h_y}\sum_{i=1}^n f^{-1}(Y_i) + \frac{1}{2n^2}\sum_{i\neq j}\frac{K_{h_y}(Y_j - Y_i) - f(Y_i)}{f(Y_i)} + O_p(h_y^{2m} + n^{-1/2}h_y^{m-1/2}).$$

By dealing with R_2 and R_3 similarly, when $nh_x^{1/2}h_y^{2m+1/2} \to 0$, $nh_x^{2m+1/2}h_y^{1/2} \to 0$, $nh_x h_y^{2m} \to 0$, $nh_x^{2m}h_y \to 0$, (A2.1) can be simplified as

$$\begin{aligned}
& n^{-1}\sum_{i=1}^n \log\left\{\frac{\widehat{f}(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{\widehat{f}(Y_i)}\right\} - n^{-1}\sum_{i=1}^n \log\left\{\frac{f(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(Y_i)}\right\} \\
= \; & \frac{\nu_2}{2n^2 h_x}\sum_{i=1}^n f^{-1}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}) + \frac{\nu_2}{2n^2 h_y}\sum_{i=1}^n f^{-1}(Y_i) - \frac{\nu_2^2}{2n^2 h_x h_y}\sum_{i=1}^n f^{-1}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i) \\
& + \frac{1}{2n}\sum_{i=1}^n \left\{1 + \frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)} - \frac{\widehat{f}(Y_i)}{f(Y_i)} - \frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}\right\} + o_p(n^{-1}h_x^{-1/2}h_y^{-1/2}).
\end{aligned}$$

Note that the remainder term is negligible uniformly for all $\boldsymbol{\beta}$. Recall that $\widehat{D}_1(\boldsymbol{\beta}) = n^{-1}\sum_{i=1}^n \left\{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)\right\}^{-1}$. With Taylor's expansion and the fact that $f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)$ is bounded away from zero and infinity, we have $\widehat{D}_1(\boldsymbol{\beta}) = n^{-1}\sum_{i=1}^n f^{-1}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i) + o_p(h_x^{1/2}h_y^{1/2})$ as $nh_x^2h_y^2 \to \infty$, $h_x^{m-1/2}h_y^{-1/2} \to 0$ and $h_x^{-1/2}h_y^{m-1/2} \to 0$. Similarly, $h_x^{-1/2}h_y^{1/2}\left[\widehat{D}_2(\boldsymbol{\beta}) - n^{-1}\sum_{i=1}^n f^{-1}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})\right]$ and $h_x^{1/2}h_y^{-1/2}\left[\widehat{D}_3 - n^{-1}\sum_{i=1}^n f^{-1}(Y_i)\right]$ are both of order $o_p(1)$. Therefore,

$$\begin{aligned}
& U_n(\boldsymbol{\beta}) - n^{-1}\sum_{i=1}^n \log\left\{\frac{f(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(Y_i)}\right\} \\
= \; & \frac{1}{2n}\sum_{i=1}^n \left\{1 + \frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)} - \frac{\widehat{f}(Y_i)}{f(Y_i)} - \frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}\right\} + o_p(n^{-1}h_x^{-1/2}h_y^{-1/2}). \quad \text{(A2.5)}
\end{aligned}$$

Next, we deal with (A2.5). We define $H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta})$ as

$$\frac{1}{2} + K_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta})K_{h_y}(Y_i - Y_j)\left\{\frac{1}{4f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)} + \frac{1}{4f(\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}, Y_j)}\right\}$$

$$-K_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}-\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta})\left\{\frac{1}{4f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}+\frac{1}{4f(\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta})}\right\}-K_{h_y}(Y_i-Y_j)\left\{\frac{1}{4f(Y_i)}+\frac{1}{4f(Y_j)}\right\}.$$

Now we show it is asymptotically a degenerate U-statistic. Because the bias of the nonparametric estimator for the last three term in $H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta})$ are of order $h_x^m + h_y^m$, h_x^m and h_y^m, respectively, we have

$$\frac{1}{n}\sum_{i=1}^{n} E\left\{H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}) \mid (\mathbf{x}_i, Y_i)\right\} - EH(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}) = O_p\left\{n^{-1/2}(h_x^m + h_y^m)\right\},$$

which is of order $o_p(n^{-1}h_x^{-1/2}h_y^{-1/2})$ when $nh_x^{2m+1}h_y \to 0$ and $nh_xh_y^{2m+1} \to 0$. In addition,

$$E\left\{\frac{K_{h_y}(Y_i-Y_j)}{f(Y_i)}\right\} = E\left[E\left\{\frac{K_{h_y}(Y_i-Y_j)}{f(Y_i)} \;\middle|\; Y_j\right\}\right] = \frac{1}{h}E\left\{\int K\left(\frac{y-Y_j}{h}\right)dy\right\} = 1,$$

leading to the fact that $E\left\{H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta})\right\} = 0$. It follows that

$$\begin{aligned}
& \frac{1}{n(n-1)}\sum_{i\neq j} H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}) \\
= \; & \frac{1}{n(n-1)}\sum_{i\neq j}\Big[H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}) - E\left\{H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}) \mid (\mathbf{x}_i, Y_i)\right\} \\
& -E\left\{H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}) \mid (\mathbf{x}_i, Y_i)\right\} + E\left\{H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta})\right\}\Big] + o_p(n^{-1}h_x^{-1/2}h_y^{-1/2}) \\
\stackrel{\text{def}}{=} \; & \frac{1}{n(n-1)}\sum_{i\neq j}\widetilde{H}(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}) + o_p(n^{-1}h_x^{-1/2}h_y^{-1/2}),
\end{aligned}$$

which is clearly a degenerate U-Statistic. This, together with (A2.5), we get that, under the null hypothesis,

$$U_n(\boldsymbol{\beta}) = \frac{1}{n(n-1)}\sum_{i\neq j}\widetilde{H}(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}) + o_p(n^{-1}h_x^{-1/2}h_y^{-1/2}). \tag{A2.6}$$

According to Theorem 2 of Major (2006), we have

$$\Pr\left(\sup_{\|\boldsymbol{\beta}\|=1} nh_x^{1/2}h_y^{1/2}\left|\frac{1}{n(n-1)}\sum_{i\neq j}\widetilde{H}(\mathbf{x}_i,Y_i,\mathbf{x}_j,Y_j;\boldsymbol{\beta})\right|\geq\epsilon\right)\leq C_1\exp\{-C_2\epsilon\}$$

with some positive constants C_1 and C_2. Take $\epsilon=\log n$, it follows that $\sup_{\|\boldsymbol{\beta}\|=1}U_n(\boldsymbol{\beta})=O_p(n^{-1}h_x^{-1/2}h_y^{-1/2}\log n)$, which proves the first assertion.

Now we turn to show that $\Pr(\widehat{\boldsymbol{\beta}}=\boldsymbol{\beta}^*)\to 1$. Since $U_n(\boldsymbol{\beta})-c_n\mathbb{I}(\boldsymbol{\beta}\neq\boldsymbol{\beta}^*)$ is maximized at $\widehat{\boldsymbol{\beta}}$, the corresponding value is naturally greater than that of a particular value $\boldsymbol{\beta}^*$, i.e., $U_n(\widehat{\boldsymbol{\beta}})-c_nI(\widehat{\boldsymbol{\beta}}\neq\boldsymbol{\beta}^*)\geq U_n(\boldsymbol{\beta}^*)$. As we have shown that both $U_n(\widehat{\boldsymbol{\beta}})$ and $U_n(\boldsymbol{\beta}^*)$ are of order $O_p(n^{-1}h_x^{-1/2}h_y^{-1/2}\log n)$, indicating that $c_nI(\widehat{\boldsymbol{\beta}}\neq\boldsymbol{\beta}^*)\leq O_p(n^{-1}h_x^{-1/2}h_y^{-1/2}\log n)$. Then when $c_nnh_x^{1/2}h_y^{1/2}/\log n\to\infty$, $I(\widehat{\boldsymbol{\beta}}\neq\boldsymbol{\beta}^*)$ is clearly of order $o_p(1)$. The proof is thus completed. □

Appendix A3: Proof of Theorem 2

According to Proposition 4 of Yin and Cook (2005), when $\boldsymbol{\beta}_0$ is unique, $\widehat{\boldsymbol{\beta}}$ converges in probability to $\boldsymbol{\beta}_0$. To study the asymptotic distribution of $\widehat{\boldsymbol{\beta}}$, it is required to study the derivative of $n^{-1}\sum_{i=1}^n\log\left\{\widehat{f}(Y_i\mid\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})\right\}$ with respect to $\boldsymbol{\beta}$. Following (A2.3), we have

$$\begin{aligned}\sup_{\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta}}\left|\widehat{f}(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta})-f(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta})\right| &= O_p\left(h_x^m+n^{-1/2}h_x^{-1/2}\log n\right),\\ \sup_{\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta},y}\left|\widehat{f}(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta},y)-f(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta},y)\right| &= O_p\left(h_x^m+h_y^m+n^{-1/2}h_x^{-1/2}h_y^{-1/2}\log n\right).\end{aligned}$$

By Taylor's expansion and condition (C3), we have

$$\log\left\{\widehat{f}(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta})\right\}-\log\{f(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta})\}-\frac{\widehat{f}(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta})-f(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta})}{f(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta})}=O_p\left(h_x^{2m}+n^{-1}h_x^{-1}\log^2 n\right)$$

holds uniformly for $\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta}$. Similarly, when the $(m-1)$th derivative of $f(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta},y)$ with

respect to $\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta}$ and y are locally Lipschitz-continuous, we have

$$\begin{aligned} & \log\left\{\widehat{f}(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta}, y)\right\} - \log\left\{f(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta}, y)\right\} - \frac{\widehat{f}(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta}, y) - f(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta}, y)}{f(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta}, y)} \\ = \; & O_p\{h_x^{2m} + h_y^{2m} + n^{-1}h_x^{-1}h_y^{-1}\log^2 n\} \end{aligned}$$

also holds uniformly for $(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{\beta}, y)$. Hence, if $nh_x^{4m} \to 0$, $nh_x^2h_y^2 \to \infty$, we have

$$\begin{aligned} & n^{-1}\sum_{i=1}^{n}\log\left\{\widehat{f}(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})\right\} - n^{-1}\sum_{i=1}^{n}\log\left\{f(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})\right\} \\ = \; & n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)} - n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})} + o_p(n^{-1/2}). \end{aligned} \tag{A3.1}$$

We now show that

$$\begin{aligned} & \frac{\partial}{\partial\boldsymbol{\beta}}\left[n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}, Y_i)} - n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}\right]\Bigg|_{\boldsymbol{\beta}=\widehat{\boldsymbol{\beta}}} \\ = \; & -n^{-1}\sum_{i=1}^{n}\left[E(\mathbf{x}_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\frac{\partial\log\left\{f(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\right\}}{\partial(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}\right] + o_p(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0 + n^{-1/2}). \end{aligned} \tag{A3.2}$$

Because the second derivative of $\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}$ with respect to $\boldsymbol{\beta}$ is $o_p(1)$ when evaluated at $\boldsymbol{\beta} = \boldsymbol{\beta}_0$, with Taylor's expansion, we get

$$\frac{\partial}{\partial\boldsymbol{\beta}}\left[n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}\right]\Bigg|_{\boldsymbol{\beta}=\widehat{\boldsymbol{\beta}}} = n^{-1}\sum_{i=1}^{n}\left[\frac{\mathbf{x}_i\widehat{f}'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} - \frac{\mathbf{x}_i f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f^2(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}\right] + o_p(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0).$$

Under condition (C4) where the m-th derivative of $f(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta})$ is locally Lipschitz-continuous with respect to $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}$, it follows that $E\{K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0) \mid \mathbf{x}_i\} = f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0) + O(h^{m-1})$ $E\{\frac{\mathbf{x}_i K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} \mid \mathbf{x}_j\} = -\frac{\partial E(\mathbf{x}_j|\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)}{\partial(\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)} + O(h^{m-1})$. In addition, because $\mathrm{var}\{K_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)\} = O(h_x^{-1})$ and $\mathrm{var}\{K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)\} = O(h_x^{-3})$, according to the projection

of U-Statistics (Serfling, 2009), we have

$$
\begin{aligned}
& n^{-1}\sum_{i=1}^{n}\frac{\mathbf{x}_i\widehat{f}'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} = \frac{1}{n(n-1)}\sum_{i\neq j}\frac{\mathbf{x}_i K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0-\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} \\
= \; & n^{-1}\sum_{i=1}^{n}\frac{\mathbf{x}_i f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} - n^{-1}\sum_{j=1}^{n}\frac{\partial E(\mathbf{x}_j \mid \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)}{\partial(\mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)} - E\left\{\frac{\mathbf{x}_i f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}\right\} + o_p(n^{-1/2}).
\end{aligned}
$$

Similarly, we have

$$
\begin{aligned}
n^{-1}\sum_{i=1}^{n}\frac{\mathbf{x}_i f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f^2(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} \; &= \; n^{-1}\sum_{i=1}^{n}\frac{\{E(\mathbf{x}_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)+\mathbf{x}_i\}\, f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} \\
& \quad -E\left\{\frac{\mathbf{x}_i f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}\right\} + o_p(n^{-1/2}).
\end{aligned}
$$

Subsequently, we know that

$$
\begin{aligned}
& \frac{\partial}{\partial\boldsymbol{\beta}}\left[n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})}\right]\Bigg|_{\boldsymbol{\beta}=\widehat{\boldsymbol{\beta}}} \\
= \; & n^{-1}\sum_{i=1}^{n}\left[\frac{E(\mathbf{x}_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} + \frac{\partial E(\mathbf{x}_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{\partial(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}\right] + o_p(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0+n^{-1/2}).
\end{aligned}
$$

Following similar arguments, we derive that

$$
\begin{aligned}
& \frac{\partial}{\partial\boldsymbol{\beta}}\left[n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta},Y_i)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta},Y_i)}\right]\Bigg|_{\boldsymbol{\beta}=\widehat{\boldsymbol{\beta}}} \\
= \; & n^{-1}\sum_{i=1}^{n}\left[\frac{E(\mathbf{x}_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0,Y_i)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0,Y_i)} + \frac{\partial E(\mathbf{x}_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{\partial(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}\right] + o_p(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0+n^{-1/2}).
\end{aligned}
$$

Then the combination of the above two equations yield (A3.2) immediately.

Recall that $\widehat{\boldsymbol{\beta}}$ is the maximizer for the following optimization problem,

$$
\begin{aligned}
\max \quad & \frac{1}{n}\sum_{i=1}^{n}\log\left\{\widehat{f}(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})\right\} + \frac{\nu_2^2}{2nh_xh_y}\widehat{D}_1(\boldsymbol{\beta}) - \frac{\nu_2}{2nh_x}\widehat{D}_2(\boldsymbol{\beta}) - c_n\mathbb{I}(\boldsymbol{\beta}\neq\boldsymbol{\beta}^*), \\
\text{s.t.} \quad & \boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{\beta}=1.
\end{aligned}
$$

We can incorporate the constraint into the objective function with the method of Lagrange multipliers (Boyd and Vandenberghe, 2004). Specifically, we maximize

$$\frac{1}{n}\sum_{i=1}^{n}\log\left\{\widehat{f}(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})\right\} + \frac{\nu_2^2}{2nh_xh_y}\widehat{D}_1(\boldsymbol{\beta}) - \frac{\nu_2}{2nh_x}\widehat{D}_2(\boldsymbol{\beta}) - c_n\mathbb{I}(\boldsymbol{\beta}\neq\boldsymbol{\beta}^*) - \lambda(\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{\beta}-1).$$

where λ is a Lagrange multiplier. When $c_n = o(n^{-1/2})$ and $nh_x^2h_y^2 \to \infty$, we take the deriverive with respect to $\boldsymbol{\beta}$,

$$\frac{\partial}{\partial\boldsymbol{\beta}}\left[n^{-1}\sum_{i=1}^{n}\log\left\{\widehat{f}(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta})\right\}\right]\Bigg|_{\boldsymbol{\beta}=\widehat{\boldsymbol{\beta}}} - 2\widehat{\boldsymbol{\beta}}\lambda = o_p(n^{-1/2}).$$

Substituting (A3.1) and (A3.2) into this equation, we can conclude that

$$\begin{aligned}
&n^{-1}\sum_{i=1}^{n}\left[\{\mathbf{x}_i - E(\mathbf{x}_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\}\,\frac{\partial\log\{f(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\}}{\partial(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}\right]\\
&+E\left[\frac{\partial^2\log\{f(Y \mid \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0)\}}{\partial\boldsymbol{\beta}_0\partial\boldsymbol{\beta}_0^{\mathrm{T}}}\right](\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0) - 2\widehat{\boldsymbol{\beta}}\lambda = o_p(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0+n^{-1/2}),
\end{aligned}$$

Furthermore, because $\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{\beta}=1$, we are able to get rid of the Lagrange multiplier λ by left multiplying $\boldsymbol{\beta}_0\boldsymbol{\beta}_0^{\mathrm{T}}$ to the equation above, and take the difference. It follows that

$$\begin{aligned}
&n^{-1}\sum_{i=1}^{n}(\mathbf{I}_p-\boldsymbol{\beta}_0\boldsymbol{\beta}_0^{\mathrm{T}})\left[\{\mathbf{x}_i - E(\mathbf{x}_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\}\,\frac{\partial\log\{f(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\}}{\partial(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}\right]\\
=\;&(\mathbf{I}_p-\boldsymbol{\beta}_0\boldsymbol{\beta}_0^{\mathrm{T}})\,E\left[\frac{\partial^2\log\{f(Y \mid \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}_0)\}}{\partial\boldsymbol{\beta}_0\partial\boldsymbol{\beta}_0^{\mathrm{T}}}\right](\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0) + o_p(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0+n^{-1/2}).
\end{aligned}$$

Thus the proof is completed. □

Appendix A4: Proof of Theorem 3

As we have shown in the proof of Theorem 1, $\Pr(\widehat{\boldsymbol{\beta}}=\boldsymbol{\beta}^*)\to 1$ under the null hypoth-

esis. Recall that we have shown (A2.6) under the null hypothesis,

$$U_n(\boldsymbol{\beta}^*) = \frac{1}{n(n-1)} \sum_{i \neq j} \widetilde{H}(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}^*) + o_p(n^{-1} h_x^{-1/2} h_y^{-1/2}).$$

We further let $G(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2)$ be defined as

$$G(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2) = E\left\{\widetilde{H}(\mathbf{x}_1, Y_1, \mathbf{x}_3, Y_3; \boldsymbol{\beta}^*)\widetilde{H}(\mathbf{x}_2, Y_2, \mathbf{x}_3, Y_3; \boldsymbol{\beta}^*) \mid (\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2)\right\}.$$

We would show that $EG^2(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2) = O(h_x^{-1} h_y^{-1})$. It suffices to observe that

$$\begin{aligned}
& E\left[E^2\left\{K_{h_y}(Y_1 - Y_3)K_{h_y}(Y_2 - Y_3) \mid (Y_1, Y_2)\right\}\right] \\
= \; & h_y^{-2} E\left[\left\{\int K(u) K\left(u + \frac{Y_1 - Y_2}{h_y}\right) f(uh_y + Y_1) du\right\}^2\right] \\
= \; & h_y^{-2} \iint \left\{\int K(u) K\left(u + \frac{y_1 - y_2}{h_y}\right) f(uh_y + y_1) du\right\}^2 f(y_1) f(y_2) dy_1 dy_2 \\
= \; & h_y^{-1} \iint \left\{\int K(u) K\left(u + v\right) f(uh_y + y_1) du\right\}^2 f(y_1) f(y_1 - h_y v) dy_1 dv \\
= \; & O(h_y^{-1}).
\end{aligned}$$

Thus, $E\left\{G^2(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2)\right\} = O(h_x^{-1} h_y^{-1})$. Next, we calculate $E\left\{\widetilde{H}^4(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2; \boldsymbol{\beta}^*)\right\}$ and show that it is of order $O(h_x^{-3} h_y^{-3})$. In fact,

$$\begin{aligned}
E\left\{K_{h_y}^4(Y_1 - Y_2))\right\} &= h_y^{-4} \int K^4\left(\frac{y_1 - y_2}{h_y}\right) f(y_1) f(y_2) dy_1 dy_2 \\
&= h_y^{-3} \int K^4(u) f(uh_y + y_2) f(y_2) du dy_2 \\
&= O(h_y^{-3}).
\end{aligned}$$

It follows similarly that $E\left\{\widetilde{H}^4(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2; \boldsymbol{\beta}^*)\right\} = O(h_x^{-3}h_y^{-3})$. Thus

$$\frac{E\left\{G^2(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2)\right\} + n^{-1}E\left\{\widetilde{H}^4(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2; \boldsymbol{\beta}^*)\right\}}{\left[E\left\{\widetilde{H}^2(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2; \boldsymbol{\beta}^*)\right\}\right]^2} \to 0$$

when $n \to \infty$ and $nh_xh_y \to \infty$. This shows that the condition in Lemma 3.2 of Zheng (1996) holds. Applying this lemma,

$$\frac{(n-1)^{-1}\sum_{i\neq j}\widetilde{H}(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j; \boldsymbol{\beta}^*)}{\left[2E\left\{\widetilde{H}^2(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2; \boldsymbol{\beta}^*)\right\}\right]^{1/2}}$$

has a limiting standard normal distribution. From the previous analysis, we know that

$$\begin{aligned}
2E\left\{H^2(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2; \boldsymbol{\beta}^*)\right\} &= E\left\{\frac{K_{h_x}^2(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}^* - \mathbf{x}_2^{\mathrm{T}}\boldsymbol{\beta}^*)}{4f(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}^*)f(\mathbf{x}_2^{\mathrm{T}}\boldsymbol{\beta}^*)}\right\}E\left\{\frac{K_{h_y}^2(Y_1 - Y_2)}{f(Y_1)f(Y_2)}\right\} \\
&\quad + E\left\{\frac{K_{h_x}^2(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}^* - \mathbf{x}_2^{\mathrm{T}}\boldsymbol{\beta}^*)K_{h_y}^2(Y_1 - Y_2)}{4f^2(\mathbf{x}_1^{\mathrm{T}}\boldsymbol{\beta}^*)f^2(Y_1)}\right\} + o(h_x^{-1}h_y^{-1}) \\
&= 2^{-1}h_x^{-1}h_y^{-1}\nu_2^2 E\left\{f^{-1}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^*, Y)\right\} + o(h_x^{-1}h_y^{-1}),
\end{aligned}$$

where $\nu_2 \stackrel{\text{def}}{=} \int K^2(u)du$. Thus with the fact that $E\left\{H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j) \mid (\mathbf{x}_i, Y_i)\right\} = O_p(h_x^m + h_y^m)$, we know $2E\left\{\widetilde{H}^2(\mathbf{x}_1, Y_1, \mathbf{x}_2, Y_2)\right\} = 2^{-1}h_x^{-1}h_y^{-1}\nu_2^2 E\left\{f^{-1}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^*, Y)\right\} + o(h_x^{-1}h_y^{-1})$. Therefore, we have $nh_x^{1/2}h_y^{1/2}U_n(\boldsymbol{\beta}^*)$ converges in distribution to a normal distribution with mean zero and variance $2^{-1}\nu_2^2 E\left\{f^{-1}(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^*, Y)\right\}$, and the proof is completed. □

Appendix A5: Proof of Theorem 4

Under the fixed alternative hypothesis, we have $\sup_{\|\boldsymbol{\beta}\|=1} I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}, Y) > 0$. This implies that there exists at least one $\boldsymbol{\beta}^\star$ such that $\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^\star$ and Y are dependent, i.e., $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^\star, Y) \geq$

$c' > 0$. Because $\widehat{\boldsymbol{\beta}}$ maximizes $U_n(\boldsymbol{\beta}) - c_n \mathbb{I}(\boldsymbol{\beta} \neq \boldsymbol{\beta}^*)$, we have

$$\begin{aligned} T_n &= U_n(\widehat{\boldsymbol{\beta}}) - c_n I(\widehat{\boldsymbol{\beta}} \neq \boldsymbol{\beta}^*) + c_n I(\widehat{\boldsymbol{\beta}} \neq \boldsymbol{\beta}^*) \\ &\geq \max_{\|\boldsymbol{\beta}\|=1} \{U_n(\boldsymbol{\beta}) - c_n \mathbb{I}(\boldsymbol{\beta} \neq \boldsymbol{\beta}^*)\} \\ &\geq U_n(\boldsymbol{\beta}^\star) - c_n \mathbb{I}(\boldsymbol{\beta}^\star \neq \boldsymbol{\beta}^*) = I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^\star, Y) + o_p(1). \end{aligned}$$

The last equation holds since $U_n(\boldsymbol{\beta}^\star)$ is a consistent estimator of $I(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta}^\star, Y)$ and $c_n \to 0$. Subsequently, $T_n \geq c' + o_p(1)$ and hence $nh_x^{1/2}h_y^{1/2}T_n \to \infty$, which completes the proof of the first assertion.

When the maximum unit direction $\boldsymbol{\beta}_0$ is unique, as we have shown in the proof of Theorem 2, $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}_0 + O_p(n^{-1/2})$. Thus

$$\begin{aligned} & \frac{1}{n}\sum_{i=1}^{n}\left\{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}) - f(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}) - \widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0) + f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\right\} \\ = & \frac{1}{n(n-1)}\sum_{i\neq j}\left\{K_{h_x}(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}} - \mathbf{x}_j^{\mathrm{T}}\widehat{\boldsymbol{\beta}}) - K_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0) + f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0) - f(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}})\right\} \\ = & \frac{(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^{\mathrm{T}}}{n(n-1)}\sum_{i\neq j}\left\{K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)(\mathbf{x}_i - \mathbf{x}_j) - f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\mathbf{x}_i\right\} + o_p(n^{-1/2}). \quad \text{(A5.1)} \end{aligned}$$

Because $\mathrm{var}\left\{K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)(\mathbf{x}_i - \mathbf{x}_j)\right\} = O(h_x^{-3})$, by using the Hoeffding decomposition (Serfling, 2009, Lemma 5.1.5A), when $nh_x^{3/2} \to \infty$, we have $\frac{1}{n(n-1)}\sum_{i\neq j} K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)(\mathbf{x}_i - \mathbf{x}_j)$ equals

$$\frac{2}{n}\sum_{i=1}^{n} E\left\{K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)(\mathbf{x}_i - \mathbf{x}_j) \mid \mathbf{x}_i\right\} - E\left\{K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)(\mathbf{x}_i - \mathbf{x}_j)\right\} + o_p(1).$$

In addition, $E\left\{K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)(\mathbf{x}_i - \mathbf{x}_j)\right\} = \frac{\partial}{\partial\boldsymbol{\beta}}\left\{EK_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta} - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta})\right\}|_{\boldsymbol{\beta}=\boldsymbol{\beta}_0}$ which equals $E\left\{f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\mathbf{x}_i\right\} + o_p(1)$. Similarly, we have $E\left\{K'_{h_x}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0 - \mathbf{x}_j^{\mathrm{T}}\boldsymbol{\beta}_0)(\mathbf{x}_i - \mathbf{x}_j) \mid \mathbf{x}_i\right\}$

equals $f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\mathbf{x}_i + o_p(1)$. Substituting these results in (A5.1), we get

$$\begin{aligned}
& \frac{1}{n}\sum_{i=1}^{n}\left\{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}) - f(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}) - \widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0) + f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\right\} \\
= \; & (\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^{\mathrm{T}}\left(\frac{1}{n}\sum_{i=1}^{n}\left[f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\mathbf{x}_i - E\left\{f'(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\mathbf{x}_i\right\}\right]\right) + o_p(n^{-1/2}),
\end{aligned}$$

which is clearly of order $o_p(n^{-1/2})$. Therefore, with Taylor's expansion, we derive that when $nh_x^{4m} \to 0$ and $nh_x^{3/2} \to \infty$,

$$\begin{aligned}
& n^{-1}\sum_{i=1}^{n}\left[\log\left\{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}})\right\} - \log\left\{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\right\}\right] \\
= \; & n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}) - f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} + o_p(n^{-1/2}) \\
= \; & n^{-1}\sum_{i=1}^{n}\frac{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0) - f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} + n^{-1}\sum_{i=1}^{n}\frac{f(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}}) - f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)} + o_p(n^{-1/2}).
\end{aligned}$$

Similar to dealing with $\frac{1}{n(n-1)}\sum_{i\neq j} H(\mathbf{x}_i, Y_i, \mathbf{x}_j, Y_j)$ in the proof of Theorem 3, the fist summation in the above display is of order $O_p(n^{-1/2}h_x^m + n^{-1}h_x^{-1/2})$. Therefore, we have

$$\begin{aligned}
& n^{-1}\sum_{i=1}^{n}\left[\log\left\{\widehat{f}(\mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}})\right\} - \log\left\{f(\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)\right\}\right] \\
= \; & \left(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\right)^{\mathrm{T}} E\left[\frac{\partial}{\partial\boldsymbol{\beta}}\log\left\{f(\mathbf{x}^{\mathrm{T}}\boldsymbol{\beta})\right\}\Big|_{\boldsymbol{\beta}=\boldsymbol{\beta}_0}\right] + o_p(n^{-1/2}).
\end{aligned}$$

Following similar arguments, we can show that as $nh_x^{3/2}h_y \to \infty$,

$$\begin{aligned}
& n^{-1}\sum_{i=1}^{n}\log\left\{\frac{\widehat{f}(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}})}{\widehat{f}(Y_i)}\right\} - n^{-1}\sum_{i=1}^{n}\log\left\{\frac{f(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(Y_i)}\right\} \\
= \; & \left(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\right)^{\mathrm{T}} E\left[\frac{\partial}{\partial\boldsymbol{\beta}}\log\left\{f(Y \mid \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta})\right\}\Big|_{\boldsymbol{\beta}=\boldsymbol{\beta}_0}\right] + o_p(n^{-1/2}).
\end{aligned}$$

By noting that $E\log\{f(Y \mid \mathbf{x}^{\mathrm{T}}\boldsymbol{\beta})\}$ is maximized at $\boldsymbol{\beta}_0$, we can easily conclude that

$$n^{-1}\sum_{i=1}^{n}\log\left\{\frac{\widehat{f}(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\widehat{\boldsymbol{\beta}})}{\widehat{f}(Y_i)}\right\} = n^{-1}\sum_{i=1}^{n}\log\left\{\frac{f(Y_i \mid \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}_0)}{f(Y_i)}\right\} + o_p(n^{-1/2}),$$

and the proof of the second assertion is completed because $\frac{\nu_2^2}{2nh_xh_y}\widehat{D}_1(\widehat{\boldsymbol{\beta}}) - \frac{\nu_2}{2nh_x}\widehat{D}_2(\widehat{\boldsymbol{\beta}}) - \frac{\nu_2}{2nh_y}\widehat{D}_3$ is of order $o_p(n^{-1/2})$ when $nh_x^2h_y^2 \to \infty$. □

Appendix A6: Proof of Theorem 5

Recall that under the local alternative, $f_{\mathbf{x},Y}(\mathbf{x},y) = f_{\mathbf{x}}(\mathbf{x})f_Y(y)\{1+\lambda_n g(\mathbf{x},y)\}$. Let $\mathbf{H}$ be any fixed p by p full rank matrix and let $\mathbf{u} = (U_1,\ldots,U_p)^{\mathrm{T}} \stackrel{\mathrm{def}}{=} \mathbf{H}\mathbf{x}$ be the transformed random vector. Then we have

$$\begin{aligned} f_{\mathbf{u},Y}(\mathbf{u},y) &= |\mathbf{H}^{-1}|f_{\mathbf{x},Y}(\mathbf{H}^{-1}\mathbf{u},y) \\ &= |\mathbf{H}^{-1}|f_{\mathbf{x}}(\mathbf{H}^{-1}\mathbf{u})f_Y(y)\{1+\lambda_n g(\mathbf{H}^{-1}\mathbf{u},y)\} \\ &= f_{\mathbf{u}}(\mathbf{u})f_Y(y)\{1+\lambda_n g(\mathbf{H}^{-1}\mathbf{u},y)\}. \end{aligned}$$

Therefore, by integrating $u_2,\ldots,u_p$ on both sides, we get

$$f_{U_1,Y}(u_1,y) = f_{U_1}(u_1)f_Y(y) + \lambda_n\int f_{\mathbf{u}}(\mathbf{u})f_Y(y)g(\mathbf{H}^{-1}\mathbf{u},y)du_2\ldots du_p.$$

Define $\widetilde{g}(u_1,y) \stackrel{\mathrm{def}}{=} f_{U_1}^{-1}(u_1)\int f_{\mathbf{u}}(\mathbf{u})g(\mathbf{H}^{-1}\mathbf{u},y)du_2\ldots du_p$. We note that there must exist some $\mathbf{H}$ such that $\widetilde{g}(u_1,y) \neq 0$ because otherwise, Y is independent of $\mathbf{x}$. Hence, we assume without loss of generality that $\widetilde{g}(u_1,y) \neq 0$. In addition,

$$\begin{aligned} & n^{-1}\sum_{i=1}^{n}\log\left\{\frac{f(Y_i \mid U_{1i})}{f(Y_i)}\right\} = n^{-1}\sum_{i=1}^{n}\log\{1+\lambda_n\widetilde{g}(U_{1i},Y_i)\} \\ = \;& \lambda_n n^{-1}\sum_{i=1}^{n}\widetilde{g}(U_{1i},Y_i) - \frac{\lambda_n^2}{2}n^{-1}\sum_{i=1}^{n}\widetilde{g}^2(U_{1i},Y_i) + o_p(\lambda_n^2) \end{aligned}$$

$$= \lambda_n E\widetilde{g}(U_1, Y) - \frac{\lambda_n^2}{2} E\widetilde{g}^2(U_1, Y) + o_p(\lambda_n^2).$$

We note that

$$\begin{aligned} E\widetilde{g}(U_1, Y) &= \iint \widetilde{g}(u_1, y) f_{U_1,Y}(u_1, y) du_1 dy \\ = & \iint f_{\mathbf{u}}(\mathbf{u}) f_Y(y) g(\mathbf{H}^{-1}\mathbf{u}, y) d\mathbf{u} dy + \lambda_n \iint \widetilde{g}^2(u_1, y) f_{U_1}(u_1) f_Y(y) du_1 dy. \end{aligned}$$

The first term is equal to zero by property 2) of $g(\cdot, \cdot)$ function. Thus, it equals $\lambda_n \Delta$, where $\Delta \stackrel{\text{def}}{=} \iint \widetilde{g}^2(u_1, y) f_{U_1}(u_1) f_Y(y) du_1 dy$ Furthermore, it can be derived that $E\widetilde{g}^2(U_1, Y) = \Delta + O(\lambda_n)$. Therefore, we have

$$n^{-1} \sum_{i=1}^{n} \log \left\{ \frac{f(Y_i \mid U_{1i})}{f(Y_i)} \right\} = 2^{-1} \lambda_n^2 \Delta + o_p(\lambda_n^2).$$

Recall (A2.5) and according to the proof of Theorem 1, it holds that

$$U_n(\mathbf{H}_1) = n^{-1} \sum_{i=1}^{n} \log \left\{ \frac{f(Y_i \mid U_{1i})}{f(Y_i)} \right\} + O_p(n^{-1} h_x^{-1/2} h_y^{-1/2}),$$

which is clearly of order $O_p(\lambda_n^2 + n^{-1} h_x^{-1/2} h_y^{-1/2})$, where $\mathbf{H}_1$ is the first column of $\mathbf{H}^{\mathrm{T}}$. Now we are ready to calculate the order of T_n under the local alternatives. Because $\widehat{\boldsymbol{\beta}}$ maximizes $U_n(\boldsymbol{\beta}) - c_n \mathbb{I}(\boldsymbol{\beta} \neq \boldsymbol{\beta}^*)$, we have

$$T_n = U_n(\widehat{\boldsymbol{\beta}}) \geq U_n(\mathbf{H}_1) - c_n I(\mathbf{H}_1 \neq \boldsymbol{\beta}^*) + c_n I(\widehat{\boldsymbol{\beta}} \neq \boldsymbol{\beta}^*) \geq U_n(\mathbf{H}_1) - c_n,$$

which implies that

$$n h_x^{1/2} h_y^{1/2} T_n \geq O_p(n h_x^{1/2} h_y^{1/2} \lambda_n^2 + 1) - c_n n h_x^{1/2} h_y^{1/2}.$$

That is, as $nh_x^{1/2}h_y^{1/2}\lambda_n^2 \to \infty$, $\lambda_n^2/c_n \to \infty$, $nh_x^{1/2}h_y^{1/2}T_n$ converges in probability to infinity, which completes the proof. $\square$

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